

Inference-Time Alignment for Diffusion Models via Doob’s Matching

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Abstract. Inference-time alignment for diffusion models aims to adapt a pre-trained diffusion model toward a target distribution without retraining the base score network, thereby preserving the generative capacity of the base model while enforcing desired properties at the inference time. A central mechanism for achieving such alignment is guidance, which modifies the sampling dynamics through an additional drift term. In this work, we introduce Doob’s matching, a novel framework for guidance estimation grounded in Doob’s h -transform. Our approach formulates guidance as the gradient of logarithm of an underlying Doob’s h -function and employs gradient-penalized regression to simultaneously estimate both the h -function and its gradient, resulting in a consistent estimator of the guidance. Theoretically, we establish non-asymptotic convergence rates for the estimated guidance. Moreover, we analyze the resulting controllable diffusion processes and prove non-asymptotic convergence guarantees for the generated distributions in the 2-Wasserstein distance.

Keywords: Controllable generative learning, inference-time alignment, Doob’s h -transform, convergence rate

1 Introduction

Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song and Ermon, 2019; Song et al., 2021) have emerged as powerful generative tools for sampling from data distributions, achieving remarkable success across diverse domains, including text-to-image and text-to-video generation (Ramesh et al., 2021), Bayesian inverse problems (Chung et al., 2023; Song et al., 2023; Chen et al., 2025; Chang et al., 2025b), and scientific applications (Bao et al., 2024; Li et al., 2025; Si and Chen, 2025; Ding et al., 2024b; Uehara et al., 2025a). Recent years have witnessed the development of large-scale diffusion models pre-trained on vast datasets. Despite the robust capabilities of these base models in capturing the training distribution, the target distributions of downstream generative tasks rarely align perfectly with this base distribution. For example, in conditional generative learning (Dhariwal and Nichol, 2021; Ho

and Salimans, 2021), base diffusion models generate samples from a mixture of distributions, whereas downstream tasks require sampling from specific constituent distributions. Similarly, in posterior sampling contexts (Chung et al., 2023; Song et al., 2023; Chen et al., 2025; Chang et al., 2025b; Purohit et al., 2025; Martin et al., 2025), the base distribution serves as the prior, while the target is the posterior defined by tilting the prior with a measurement likelihood. Furthermore, downstream tasks frequently impose additional constraints, such as human preferences or safety considerations (Domingo-Enrich et al., 2024, 2025; Uehara et al., 2025b; Kim et al., 2025; Sabour et al., 2025; Denker et al., 2025; Ren et al., 2025), which must be satisfied without compromising the model’s generative quality.

To bridge the gap between base and target distributions, researchers have proposed numerous alignment methods (Xu et al., 2023; Lee et al., 2023b; Fan et al., 2023; Clark et al., 2024; Domingo-Enrich et al., 2024, 2025; Uehara et al., 2024b, 2025b). These strategies generally fall into two categories: fine-tuning and inference-time alignment. Fine-tuning approaches involve retraining the base score network via supervised learning (Lee et al., 2023b), reinforcement learning (Fan et al., 2023; Black et al., 2024; Clark et al., 2024; Uehara et al., 2024c), or classifier-free fine-tuning (Ho and Salimans, 2021; Zhang et al., 2023; Yuan et al., 2023). Despite its conceptual simplicity, fine-tuning presents significant limitations. First, it often requires a substantial collection of high-quality samples from the target distribution, which may be unavailable in practical scenarios like posterior sampling. Second, the computational cost of retraining score networks can be prohibitive, particularly for large-scale models with billions of parameters (Uehara et al., 2025b). Third, fine-tuning is vulnerable to over-optimization (Gao et al., 2023; Rafailov et al., 2024; Kim et al., 2025), where the network overfits to limited target samples or preferences, causing it to “forget” the valuable prior information encoded in the base model. This degradation undermines the fundamental advantage of leveraging pre-trained models.

In contrast, inference-time alignment (Uehara et al., 2025b; Kim et al., 2025; Sabour et al., 2025; Denker et al., 2025; Ren et al., 2025; Patchebbat et al., 2025) eliminates the need to retrain the base diffusion model. These methods offer substantial computational advantages and preserve the generative capacity of the underlying model. The core technique is guidance (Jiao et al., 2025), which incorporates target information by introducing an additional drift term to the base diffusion model. Within the framework of Doob’s h -transform (Rogers and Williams, 2000; Särkkä and Solin, 2019; Chewi, 2025), the score function for the target tilted distribution decomposes into the sum of the base score and a guidance term, where the guidance is defined as the gradient of the log-Doob’s h -function (Heng et al., 2024; Tang and Xu, 2024; Denker et al., 2024, 2025). This relationship has also been investigated through the lens of classifier guidance (Dhariwal and Nichol, 2021), stochastic optimal control (Han et al., 2024; Tang and Zhou, 2025), and Bayes’ rule (Chung et al., 2023; Song et al., 2023). The primary challenge lies in accurately estimating this guidance.

Guidance estimation methods can be categorized into two main approaches: approximation and learning. The approximation approach, exemplified by diffusion posterior sampling (Chung et al., 2023) and loss-guided diffusion (Song et al., 2023), relies on heuristic approximations that often lead to inconsistencies with the underlying mathematical formulation. The learning approach (Dhariwal and Nichol, 2021; Tang and Xu, 2024; Denker et al., 2024, 2025) attempts to learn the necessary components by deep neural networks. Classifier guidance (Dhariwal and Nichol, 2021) learns Doob’s h -function via a classifier, but this is effective primarily for discrete labels. Furthermore, the convergence of the plug-in gradient estimator of the classifier is not guaranteed, potentially undermining guidance reliability (Mou,

2025, Section 3.2.2). To mitigate this, Tang and Xu (2024) estimate Doob’s h -function and its gradient using separate neural networks, increasing training complexity. While Denker et al. (2024) directly learn the guidance, their method requires samples from the target tilted distribution. Denker et al. (2025) attempt to address this data requirement via iterative retraining.

To address these limitations, we introduce Doob’s h -matching, a novel framework for provable guidance estimation in inference-time alignment. We propose a gradient-penalized regression method that simultaneously estimates Doob’s h -function and its gradient, yielding a consistent estimator of the guidance. When combined with the pre-trained base score, our method enables efficient sampling from the target distribution without the need for computationally expensive fine-tuning or access to target distribution samples.

1.1 Contributions. Our main contributions are summarized as follows:

- (i) We introduce *Doob’s matching*, a novel guidance estimation framework for controllable diffusion models grounded in Doob’s h -transform. The Doob h -function is estimated via a least-squares regression approach augmented with a gradient penalty, and the gradient of the logarithm of the resulting h -function estimator yields an estimator for the Doob’s guidance. Additionally, this method is derivative-free, meaning it does not require access to the gradient of the weight function between the target tilted distribution and the base distribution.
- (ii) We establish non-asymptotic convergence rates for Doob’s matching, showing that the proposed method guarantees convergence of both the h -function estimator and its gradient. These results provide rigorous theoretical guarantees for Doob’s guidance estimation (Theorem 4.3). Moreover, we derive non-asymptotic convergence rates for the induced controllable diffusion models, thereby establishing rigorous guarantees for the generated distributions in the 2-Wasserstein distance (Theorem 4.5).

1.2 Organization. The remainder of this paper is organized as follows. In Section 2, we provide a brief introduction to diffusion models. In Section 3, we propose the stochastic dynamics of controllable diffusion models within the framework of Doob’s h -transform, and we present a practical algorithm to simultaneously estimate Doob’s h -function and its gradient. Section 4 establishes non-asymptotic error bounds for both the estimation of the h -function and the induced controllable diffusion models. Finally, concluding remarks are provided in Section 5. Detailed proofs of theoretical results are deferred to the appendix.

2 Preliminaries on Diffusion Models

2.1 Forward and time-reversal process. We consider the diffusion model for a base distribution p_0 . The forward process of the base diffusion model is defined by the Ornstein–Uhlenbeck process:

$$(2.1) \quad d\mathbf{X}_t = -\mathbf{X}_t dt + \sqrt{2} d\mathbf{B}_t, \quad t \in (0, T), \quad \mathbf{X}_0 \sim p_0,$$

where $\mathbf{B}_t$ is a d -dimensional standard Brownian motion, and $T > 0$ is the terminal time. The transition distribution of the forward process can be expressed as:

$$(2.2) \quad (\mathbf{X}_t | \mathbf{X}_0 = \mathbf{x}_0) \sim \mathcal{N}(\mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d),$$

where the mean and variance coefficients are given, respectively, as $\mu_t = \exp(-t)$ and $\sigma_t^2 = 1 - \exp(-2t)$. The forward process (2.1) is commonly referred to as the variance-preserving (VP) SDE (Song et al., 2021) as $\mu_t^2 + \sigma_t^2 = 1$ for each $t \in (0, T)$. Denote by p_t the

marginal density of $\mathbf{X}_t$ for each $t \in (0, T)$, which satisfies

$$p_t(\mathbf{x}_t) = \int \varphi_d(\mathbf{x}_t; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0,$$

where $\varphi_d(\cdot; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d)$ denotes the density function of the Gaussian distribution $\mathcal{N}(\mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d)$. The corresponding time-reversal process (Anderson, 1982) of (2.1) is characterized by:

$$(2.3) \quad \begin{aligned} d\mathbf{X}_t^\leftarrow &= \left(\mathbf{X}_t^\leftarrow + 2 \overbrace{\nabla \log p_{T-t}(\mathbf{X}_t^\leftarrow)}^{\text{base score}} \right) dt + \sqrt{2} d\mathbf{B}_t, \quad t \in (0, T), \\ \mathbf{X}_0^\leftarrow &\sim p_T, \end{aligned}$$

where the marginal density of $\mathbf{X}_t^\leftarrow$ coincides p_{T-t} for each $t \in (0, T)$.

2.2 Path measure and filtration. We formally define the probability space for the time-reversal process (2.3). Let $\Omega := C([0, T], \mathbb{R}^d)$ be the space of continuous functions mapping $[0, T]$ to $\mathbb{R}^d$, equipped with the topology of uniform convergence. Let $\mathcal{F}$ be the Borel σ -algebra on Ω . We define the canonical process $\mathbf{X}^\leftarrow$ on Ω via the coordinate mapping $\mathbf{X}_t^\leftarrow(\omega) = \omega(t)$ for all $\omega \in \Omega$. The natural filtration is given by $\mathbb{F} := (\mathcal{F}_t)_{0 \leq t \leq T}$, where $\mathcal{F}_t := \sigma(\mathbf{X}_s^\leftarrow \mid 0 \leq s \leq t)$ is the σ -algebra generated by the path up to time t . We denote by $\mathbb{P}$ the probability measure on $(\Omega, \mathcal{F})$ induced by the law of the solution to the SDE (2.3) with initial distribution $\mathbf{X}_0^\leftarrow \sim p_T$. Consequently, the filtered probability space is denoted as $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, and $\mathbf{B}_t$ is an $\mathbb{F}$ -Brownian motion under $\mathbb{P}$.

2.3 Training phase: base score matching. In generative learning, the exact base score $\nabla \log p_t$ in the time-reversal process (2.3) is intractable. One can estimate the base score using samples from the base density p_0 via standard techniques such as implicit score matching (Hyvärinen, 2005), sliced score matching (Song et al., 2020), and denoising score matching (Vincent, 2011). Let $\hat{\mathbf{s}} : (0, T) \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ denote an estimator for the prior score, that is,

$$(2.4) \quad \|\hat{\mathbf{s}}(t, \cdot) - \nabla \log p_{T-t}\|_{L^2(p_{T-t})} \leq \varepsilon,$$

for a small tolerance $\varepsilon \ll 1$. Considerable research has established theoretical guarantees for this score estimation (Tang and Yang, 2024; Oko et al., 2023; Fu et al., 2024; Ding et al., 2024a), leveraging standard techniques from non-parametric regression with deep neural networks (Bauer and Kohler, 2019; Schmidt-Hieber, 2020; Kohler and Langer, 2021; Jiao et al., 2023).

2.4 Inference phase: sampling. Given a base score estimator $\hat{\mathbf{s}}$ in (2.4), the inference phase of diffusion models aims to generate samples by simulating the time-reversal process with estimated score. Since the explicit solution of the time-reversal process is intractable, we employ an exponential integrator (Hochbruck and Ostermann, 2005, 2010; Lu et al., 2022a; Zhang and Chen, 2023). This approach is well-suited for solving the time-reversal process due to the semi-linearity of the drift term of the SDE in (2.3).

Let $K \in \mathbb{N}$ denote the number of discretization steps, and let $T_0 > 0$ be an early-stopping time. We define a sequence of uniform time points $t_k := kh$ for $k = 0, \dots, K$, where the step size is $h := (T - T_0)/K$. In each time sub-interval, the exponential integrator approximates the score function by its value at the left endpoint:

$$(2.5) \quad \begin{aligned} d\hat{\mathbf{X}}_t^\leftarrow &= (\hat{\mathbf{X}}_t^\leftarrow + 2\hat{\mathbf{s}}(kh, \hat{\mathbf{X}}_{kh}^\leftarrow)) dt + \sqrt{2} d\mathbf{B}_t, \quad t \in [kh, (k+1)h), \\ \hat{\mathbf{X}}_0^\leftarrow &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \end{aligned}$$

where $0 \leq k \leq K - 1$. The resulting linear approximation to the original semi-linear SDE has the following explicit solution:

$$\widehat{\mathbf{X}}_{(k+1)h}^{\leftarrow} = \exp(h)\widehat{\mathbf{X}}_{kh}^{\leftarrow} + 2\phi^2(h)\widehat{\mathbf{s}}(kh, \widehat{\mathbf{X}}_{kh}^{\leftarrow}) + \phi(2h)\boldsymbol{\xi}_k, \quad 0 \leq k \leq K - 1,$$

where $\phi(z) := \sqrt{\exp(z) - 1}$, and $\boldsymbol{\xi}_0, \dots, \boldsymbol{\xi}_{K-1}$ are i.i.d. standard Gaussian random variables.

Remark 2.1 (Initialization). Note that the true initial distribution of the time-reversal process (2.3) is p_T , rather than the $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ used in (2.5). We adopt the standard normal distribution because sampling from $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ is significantly more computationally tractable. This approximation is justified by the fact that p_T converges to $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ exponentially in KL-divergence as $T \rightarrow \infty$ (Bakry et al., 2014; Vempala and Wibisono, 2019); thus, the Gaussian initialization is valid for a sufficiently large terminal time T .

3 Controllable Diffusion Models via Doob's Matching

In this section, we propose controllable diffusion models for sampling from a target distribution, defined as the base distribution tilted by a weight function. We utilize the theory of measure change for diffusion processes on the filtered space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$.

3.1 Problem setup. We assume access to a pre-trained base diffusion model (2.5) that generates samples approximately from the base distribution p_0 . We aim to sample from a tilted distribution q_0 , defined by reweighting the base distribution with a known weight function $w : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$:

$$(3.1) \quad q_0(\mathbf{x}) := \frac{w(\mathbf{x})p_0(\mathbf{x})}{Z}, \quad \text{where } Z := \int w(\mathbf{x})p_0(\mathbf{x}) \, d\mathbf{x} < \infty.$$

Our goal is to derive a new diffusion process that generates samples from the tilted distribution q_0 directly by introducing a drift correction to the pre-trained base diffusion model (2.5).

This problem encompasses a wide range of application scenarios.

Example 1 (Bayesian inverse problems). Bayesian inverse problems play a critical role in scientific computing (Stuart, 2010; Kantas et al., 2014; Ding et al., 2024b), image science (Chung et al., 2023; Mardani et al., 2024; Purohit et al., 2025; Chang et al., 2025b), and medical imaging (Song et al., 2022). In Bayesian inverse problems, we aim to recover an unknown signal $\mathbf{X}_0 \in \mathbb{R}^d$ from noisy measurements $\mathbf{Y} \in \mathbb{R}^m$, which are linked by the following measurement model:

$$(3.2) \quad \mathbf{Y} = \mathcal{A}(\mathbf{X}_0) + \mathbf{n},$$

where $\mathcal{A} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ is a known measurement operator, and $\mathbf{n} \in \mathbb{R}^m$ represents a measurement noise with a known distribution. The Bayesian approach incorporates prior knowledge about $\mathbf{X}_0$ in the form of a prior distribution p_0 . Given observed measurements $\mathbf{Y} = \mathbf{y}$, the goal of Bayesian inverse problems is to sample from the posterior distribution:

$$(3.3) \quad q_0(\mathbf{x}) := p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = \frac{w(\mathbf{x})p_0(\mathbf{x})}{Z},$$

where $w(\mathbf{x}) := p_{\mathbf{Y}|\mathbf{X}_0}(\mathbf{y}|\mathbf{x})$ is a likelihood determined by the measurement model (3.2), and Z is a partition function to ensure q_0 is a valid probability density. For example, for a Gaussian noise $\mathbf{n} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, it holds that

$$w(\mathbf{x}) = (2\pi\sigma^2)^{-\frac{d}{2}} \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{y} - \mathcal{A}(\mathbf{x})\|_2^2\right).$$

In Bayesian inverse problems, one typically has a base diffusion model pre-trained on the prior distribution, and aims to sample from the posterior distribution (3.3) without retraining the base model.

Example 2 (Reward-guided generation). In the reward-guided generation (Domingo-Enrich et al., 2025; Uehara et al., 2025b; Kim et al., 2025; Sabour et al., 2025; Denker et al., 2025; Ren et al., 2025), human preferences and constraints can be encoded into a reward function $r : \mathbb{R}^d \rightarrow \mathbb{R}$. For instance, in text-to-image generation, the reward function r quantifies how well the generated data aligns with the input prompt. In practice, such reward function can be learned from the human feedback or preference data (Stiennon et al., 2020; Ouyang et al., 2022; Lee et al., 2023b). For the sake of simplicity, we assume throughout this work that the reward function has already been given. A naive approach to reward-guided generation is to maximize the expected reward $\max_{\pi \in \mathcal{P}} \mathbb{E}_{\pi}[r]$, where $\mathcal{P}$ is the set of probability measures on $\mathbb{R}^d$. However, solely maximizing the expected reward may lead to over-optimization and degenerate solutions (Kim et al., 2025). To mitigate this, entropy regularization (Uehara et al., 2024a; Tang, 2024) is incorporated into the objective functional, yielding the following optimization problem:

$$(3.4) \quad q_0 := \arg \max_{\pi \in \mathcal{P}} \mathbb{E}_{\pi}[r] - \alpha \text{KL}(\pi \| p_0),$$

where $\alpha > 0$ is a regularization parameter, and p_0 is the density of a reference distribution, i.e., the distribution corresponding to the pre-trained base diffusion model. This objective comprises two components: the expected reward, which captures human preferences, and the KL-divergence term, which prevents the distribution from deviating excessively from the base diffusion model. The closed-form solution to this optimization problem (3.4) is given by (Rafailov et al., 2023):

$$q_0(\mathbf{x}) = \frac{w(\mathbf{x})p_0(\mathbf{x})}{Z}, \quad w(\mathbf{x}) := \exp\left(\frac{r(\mathbf{x})}{\alpha}\right),$$

where Z is the partition function to ensure q_0 is a valid probability density. In reward-guided generation, the central objective is to incorporate preferences exclusively during the inference phase, thereby avoiding the substantial computational cost of retraining the large-scale base diffusion model.

Example 3 (Transfer learning for diffusion models). Diffusion models have achieved remarkable success in image generation. However, their performance critically depends on the availability of large-scale training data. In the scenarios of few-shot generation, training diffusion models solely on limited samples typically results in poor generative performance. To retain strong generative capability under data scarcity, a common strategy is to transfer expressive diffusion models pre-trained on large datasets to the target domain (Ouyang et al., 2024; Wang et al., 2024; Zhong et al., 2025; Bahram et al., 2026). Formally, transfer learning aims to adapt a model pre-trained on a large-scale source distribution to a target distribution of much smaller size and diversity of samples. However, directly fine-tuning a large pre-trained diffusion model using only limited target samples often leads to severe overfitting (Wang et al., 2024). To address this issue, transfer learning approaches for diffusion models typically train a lightweight guidance network on the limited target data and combine it with the pre-trained base score network, yielding a modified diffusion model capable of sampling from the target distribution (Ouyang et al., 2024; Zhong et al., 2025; Bahram et al., 2026). Concretely, Ouyang

et al. (2024) estimates the density ratio between the target and source distributions,

$$w(\mathbf{x}) := \frac{q_0(\mathbf{x})}{p_0(\mathbf{x})},$$

using limited samples from the target distribution. For simplicity, we assume throughout this work that the density ratio is known. Under this assumption, the objective of transfer learning for diffusion models is to sample from the target distribution q_0 by leveraging the pre-trained base diffusion model together with the estimated density ratio.

3.2 Controllable diffusion models with Doob's h -transform. In this subsection, we achieve inference-time alignment by incorporating guidance into the base diffusion models via Doob's h -transform (Rogers and Williams, 2000; Särkkä and Solin, 2019; Heng et al., 2024; Chewi, 2025). We begin by constructing a target path measure $\mathbb{Q}$ on the filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We assume that $\mathbb{Q}$ is absolutely continuous with respect to the base path measure $\mathbb{P}$ defined in Section (2.2). By the Radon-Nikodym theorem, this relationship is characterized by the existence of an $\mathbb{F}$ -adapted process $(L_t)_{0 \leq t \leq T}$ with $L_t \geq 0$, such that for every $t \in [0, T]$:

$$(3.5) \quad L_t := \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}.$$

The process L_t is known as the Radon-Nikodym derivative process, representing the likelihood ratio between the target and base measures conditioned on the filtration $\mathcal{F}_t$.

We impose two boundary conditions on the target path measure $\mathbb{Q}$: (i) the marginal distribution of the initial state $\mathbf{X}_0^\leftarrow$ under $\mathbb{Q}$ must match that of $\mathbb{P}$; and (ii) the marginal distribution of the terminal state $\mathbf{X}_T^\leftarrow$ under $\mathbb{Q}$ must coincide with the tilted distribution q_0 defined in (3.1). To enforce these constraints, we specify the Radon-Nikodym derivatives of $\mathbb{Q}$ with respect to $\mathbb{P}$ at the initial and terminal times as follows:

$$(3.6) \quad L_0 = \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_0} \equiv 1, \quad \text{and} \quad L_T = \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_T} = \frac{d\mathbb{Q}}{d\mathbb{P}} = \frac{w(\mathbf{X}_T^\leftarrow)}{\mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow)]},$$

where the denominator serves as the normalizing constant ensuring that $\mathbb{Q}$ is a valid probability measure, i.e., $\mathbb{E}^\mathbb{P}[L_T] = 1$.

The rest of our derivation proceeds in two steps. First, we characterize the stochastic dynamics of $\mathbf{X}_t^\leftarrow$ under the target measure $\mathbb{Q}$. Second, by invoking the weak uniqueness of solutions to stochastic differential equations (Øksendal, 2003, Lemma 5.3.1), we construct a controllable diffusion process under the base path measure $\mathbb{P}$ whose terminal distribution coincides with the target tilted distribution q_0 .

Dynamics under the target path measure. Girsanov's theorem (Øksendal, 2003, Theorem 8.6.8) establishes a fundamental correspondence between a drift shift in the driving Brownian motion and the dynamics of the associated Radon-Nikodym derivative process. Accordingly, we first characterize the evolution of the time-dependent likelihood ratio L_t .

Proposition 3.1 (Doob's h -function). *The Radon-Nikodym derivative process L_t , as defined in (3.5), admits the following representation:*

$$L_t = \mathbb{E}^\mathbb{P} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} \right] = \frac{h^*(t, \mathbf{X}_t^\leftarrow)}{\mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow)]},$$

where $h^* : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$, referred to as Doob's h -function, is defined as the conditional expectation of the terminal weight:

$$(3.7) \quad h^*(t, \mathbf{x}) := \mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow}) \mid \mathbf{X}_t^{\leftarrow} = \mathbf{x}].$$

Furthermore, the log-likelihood ratio satisfies the following SDE:

$$d(\log L_t) = \nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})^\top \sqrt{2} d\mathbf{B}_t - \|\nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})\|_2^2 dt.$$

The proof of Proposition 3.1 is provided in Appendix B. With the dynamics of $\log L_t$ established, we derive the stochastic dynamics of $\mathbf{X}_t^{\leftarrow}$ under the target measure $\mathbb{Q}$ via Girsanov's theorem (Øksendal, 2003, Theorem 8.6.6).

Proposition 3.2. *Let the base process $\mathbf{X}_t^{\leftarrow}$ satisfy the SDE (2.3) under the path measure $\mathbb{P}$, and let $\mathbb{Q}$ be the target path measure defined by (3.6). Assume that the Novikov condition holds:*

$$(3.8) \quad \mathbb{E}^{\mathbb{P}} \left[\exp \left(\int_0^T \|\nabla \log h^*(s, \mathbf{X}_s^{\leftarrow})\|_2^2 ds \right) \right] < \infty.$$

Define a process $(\tilde{\mathbf{B}}_t)_{1 \leq t \leq T}$ by

$$(3.9) \quad d\tilde{\mathbf{B}}_t = d\mathbf{B}_t - \sqrt{2} \nabla \log h^*(t, \mathbf{X}_t^{\leftarrow}) dt,$$

where $\mathbf{B}_t$ is a Brownian motion under $\mathbb{P}$. Then $\tilde{\mathbf{B}}_t$ is a standard Brownian motion under $\mathbb{Q}$. Further, under the path measure $\mathbb{Q}$, the base process $\mathbf{X}_t^{\leftarrow}$ in (2.3) evolves according to:

$$(3.10) \quad d\mathbf{X}_t^{\leftarrow} = (\mathbf{X}_t^{\leftarrow} + 2\nabla \log p_{T-t}(\mathbf{X}_t^{\leftarrow}) + 2\nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})) dt + \sqrt{2} d\tilde{\mathbf{B}}_t.$$

The proof is deferred to Appendix B. By the construction in (3.6), the law of $\mathbf{X}_T^{\leftarrow}$ under $\mathbb{Q}$ coincides with the target tilted distribution q_0 .

Remark 3.3 (Novikov condition). The condition in (3.8) ensures that the exponential local martingale defined by the drift shift is a true martingale (Karatzas and Shreve, 1998, Corollary 5.13), which is sufficient for the Radon-Nikodym derivative to be well-defined and for Girsanov's theorem to apply.

Controllable diffusion process under the base measure. Although Proposition 3.2 constructs stochastic dynamics driven by Brownian motion under $\mathbb{Q}$ that achieve the desired terminal distribution q_0 , practical implementation necessitates an SDE driven by Brownian motion under the reference measure $\mathbb{P}$.

To address this, we construct a surrogate process $\mathbf{Z}_t^{\leftarrow}$ on the base probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ that adopts the drift derived for $\mathbb{Q}$:

$$(3.11) \quad d\mathbf{Z}_t^{\leftarrow} = \left(\underbrace{\mathbf{Z}_t^{\leftarrow} + 2\nabla \log p_{T-t}(\mathbf{Z}_t^{\leftarrow})}_{\text{base score}} + \underbrace{2\nabla \log h^*(t, \mathbf{Z}_t^{\leftarrow})}_{\text{Doob's guidance}} \right) dt + \sqrt{2} d\mathbf{B}_t.$$

Since the process $\mathbf{Z}_t^{\leftarrow}$ driven by the $\mathbb{P}$ -Brownian motion (3.11) satisfies the same SDE as $\mathbf{X}_t^{\leftarrow}$ driven by the $\mathbb{Q}$ -Brownian motion (3.10), the weak uniqueness property of SDE solutions guarantees (Øksendal, 2003, Lemma 5.3.1) that the law of $\mathbf{Z}_t^{\leftarrow}$ under $\mathbb{P}$ is identical to the law of $\mathbf{X}_t^{\leftarrow}$ under $\mathbb{Q}$ for all $t \in [0, T]$. Thus, the law of $\mathbf{Z}_T^{\leftarrow}$ under $\mathbb{P}$ coincides the target tilted distribution q_0 . For a detailed formal statement, see Øksendal (2003, Theorem 8.6.8).

3.3 Doob's matching with gradient penalty. Up to now, we have found a controllable diffusion process (3.11) that can generate samples from the target tilted distribution q_0 . However, the Doob's h -function defined as (3.7) is intractable. This subsection aims to estimate the Doob's guidance $\nabla \log h_t^*$ in (3.11).

For every $t \in (0, T)$, the Doob's h -function $h_t^* := h^*(t, \cdot)$ is the unique solution to the following optimization problem:

$$(3.12) \quad \min_{h_t: \mathbb{R}^d \rightarrow \mathbb{R}} \mathbb{E}^{\mathbb{P}} [\|h_t(\mathbf{X}_t^{\leftarrow}) - h_t^*(\mathbf{X}_t^{\leftarrow})\|_2^2] = \|h_t - h_t^*\|_{L^2(p_{T-t})}^2.$$

However, since this objective functional includes unknown function h_t^* , we cannot use this explicit Doob's matching objective functional. Based on the property of the conditional expectation (3.7), we now propose an alternative objective functional, named implicit Doob's matching:

$$(3.13) \quad \begin{aligned} \mathcal{J}_t(h_t) &= \mathbb{E}^{\mathbb{P}} [\|h_t(\mathbf{X}_t^{\leftarrow}) - w(\mathbf{X}_T^{\leftarrow})\|_2^2] \\ &= \mathbb{E}_{\mathbf{X}_0 \sim p_0} \mathbb{E}_{\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} [\|h_t(\mu_{T-t}\mathbf{X}_0 + \sigma_{T-t}\boldsymbol{\varepsilon}) - w(\mathbf{X}_0)\|_2^2], \end{aligned}$$

where $w: \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ is the known weight function defined as (3.1).

The following proposition bridges the objective functionals of the explicit Doob's matching (3.12) and the implicit Doob's matching (3.13).

Proposition 3.4. *For every $t \in (0, T)$, the Doob's h -function h_t^* in (3.7) minimizes the implicit Doob's matching loss (3.13). Further,*

$$\mathcal{J}_t(h_t) = \mathbb{E}^{\mathbb{P}} [\|h_t(\mathbf{X}_t^{\leftarrow}) - h_t^*(\mathbf{X}_t^{\leftarrow})\|_2^2] + V_t^2,$$

where $V_t^2 := \mathbb{E}^{\mathbb{P}} [\text{Var}(w(\mathbf{X}_T^{\leftarrow}) | \mathbf{X}_t^{\leftarrow})]$ is a constant independent of h_t .

The proof of Proposition 3.4 can be found in Appendix B. Since the expectation in the population risk (3.13) is computationally intractable, we approximate it by empirical risk using independent and identically distributed samples:

$$(3.14) \quad \hat{\mathcal{J}}_t(h_t) := \frac{1}{n} \sum_{i=1}^n \|h_t(\mu_{T-t}\mathbf{X}_0^i + \sigma_{T-t}\boldsymbol{\varepsilon}^i) - w(\mathbf{X}_0^i)\|_2^2,$$

where $\mathbf{X}_0^1, \dots, \mathbf{X}_0^n$ are independent and identically distributed random variables drawn from the base distribution p_0 , and $\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_n$ are independent standard Gaussian random variables. A naive approach for estimating the Doob's h -function h_t^* (3.7) involves vanilla empirical risk minimization:

$$(3.15) \quad \hat{h}_t \in \arg \min_{h_t \in \mathcal{H}_t} \hat{\mathcal{J}}_t(h_t),$$

where $\mathcal{H}_t$ is a hypothesis class, which is chosen as a neural network class in this work.

Limitations of vanilla regression. It is crucial that estimating the Doob's guidance $\nabla \log h_t^*$ in (3.11) necessitates not only the estimation of the Doob's h -function h_t^* but also its gradient ∇h_t^* , as expressed by

$$\nabla \log h_t^*(\mathbf{z}) = \frac{\nabla h_t^*(\mathbf{z})}{h_t^*(\mathbf{z})}, \quad \mathbf{z} \in \mathbb{R}^d.$$

Previous research, including classifier guidance (Dhariwal and Nichol, 2021), Monte Carlo regression Uehara et al. (2025b, Section 2.2) and Ouyang et al. (2024), approaches this

challenge by approximating the Doob's h -function h_t^* with the vanilla empirical risk minimizer $\hat{h}_t$ (3.15), and subsequently approximating the gradient of the Doob's h -function ∇h_t^* via the plug-in gradient estimator $\nabla \hat{h}_t$.

Although rigorous theoretical studies (Bauer and Kohler, 2019; Nakada and Imaizumi, 2020; Schmidt-Hieber, 2020; Kohler and Langer, 2021; Farrell et al., 2021; Kohler et al., 2022; Jiao et al., 2023) demonstrate that the vanilla empirical risk minimizer $\hat{h}_t$ converges to the Doob's h -function h_t^* in the L^2 -norm, the convergence of the corresponding plug-in gradient estimator $\nabla \hat{h}_t$ cannot be guaranteed, which has been mentioned in Mou (2025, Section 3.2.2). To illustrate this fundamental limitation, consider the sequence of functions $f_n : \mathcal{I} \rightarrow \mathbb{R}$, defined as $x \mapsto n^{-1} \sin(nx)$, and the zero function $f_0 : \mathcal{I} \rightarrow \mathbb{R}$, where $\mathcal{I} := [0, 2\pi]$. While $\|f_n - f_0\|_{L^p(\mathcal{I})} \rightarrow 0$ as $n \rightarrow \infty$, we observe that $\lim_{n \rightarrow \infty} \|f'_n - f'_0\|_{L^p(\mathcal{I})} \neq 0$ for each $1 \leq p \leq \infty$. Consequently, the convergence of the plug-in Doob's guidance estimator $\nabla \log \hat{h}_t$ used by Uehara et al. (2025b); Ouyang et al. (2024) cannot be guaranteed in general.

To address this issue, Tang and Xu (2024) estimates the Doob's h -function and its gradient separately through a martingale approach. In contrast, this work proposes an approach that simultaneously estimate the Doob's h -function and its gradient.

Doob's matching with gradient penalty. To simultaneously estimate the Doob's h -function and its gradient, we adopt gradient-penalized regression (Drucker and Le Cun, 1991, 1992; Ding et al., 2025). The population risk is defined by incorporating an additional Sobolev regularization to (3.13):

$$(3.16) \quad \begin{aligned} \mathcal{J}_t^\lambda(h_t) &:= \mathcal{J}_t(h_t) + \overbrace{\lambda \mathbb{E}^{\mathbf{P}} [\|\nabla h_t(\mathbf{X}_t^{\leftarrow})\|_2^2]}^{\text{gradient penalty}} \\ &= \mathcal{J}_t(h_t) + \lambda \mathbb{E}_{\mathbf{X}_0 \sim p_0} \mathbb{E}_{\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} [\|\nabla h_t(\mu_{T-t} \mathbf{X}_0 + \sigma_{T-t} \boldsymbol{\varepsilon})\|_2^2], \end{aligned}$$

where $\lambda > 0$ is a regularization parameter.

By a similar argument as (3.14), we can also introduce an empirical counterpart of (3.16):

$$(3.17) \quad \hat{\mathcal{J}}_t^\lambda(h_t) := \hat{\mathcal{J}}_t(h_t) + \frac{\lambda}{n} \sum_{i=1}^n \|\nabla h_t(\mu_{T-t} \mathbf{X}_0^i + \sigma_{T-t} \boldsymbol{\varepsilon}^i)\|_2^2,$$

where $\mathbf{X}_0^1, \dots, \mathbf{X}_0^n$ are independent and identically distributed random variables random variables drawn from the base distribution p_0 , and $\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_n$ are independent standard Gaussian random variables. Then we have a gradient penalized empirical risk minimizer:

$$(3.18) \quad \hat{h}_t^\lambda \in \arg \min_{h_t \in \mathcal{H}_t} \hat{\mathcal{J}}_t^\lambda(h_t),$$

where $\mathcal{H}_t$ is a hypothesis class, which is chosen as a neural network class in this work.

Remark 3.5. In Section 4.3, we will demonstrate that not only does the gradient-penalized estimator $\hat{h}_t^\lambda$ in (3.18) converges to the Doob's h -function h_t^* , but its gradient $\nabla \hat{h}_t^\lambda$ also approaches h_t^* , provided that the regularization parameter λ and the hypothesis class $\mathcal{H}_t$ are chosen appropriately.

Doob's guidance estimation. The Doob's matching with gradient penalty (3.18) yields a valid estimator of the Doob's guidance:

$$(3.19) \quad \hat{\mathbf{g}}_t^\lambda(\mathbf{z}) := \nabla \log \hat{h}_t^\lambda(\mathbf{z}) = \frac{\nabla \hat{h}_t^\lambda(\mathbf{z})}{\hat{h}_t^\lambda(\mathbf{z})} \approx \frac{\nabla h_t^*(\mathbf{z})}{h_t^*(\mathbf{z})} = \nabla \log h_t^*(\mathbf{z}), \quad \mathbf{z} \in \mathbb{R}^d.$$

This plug-in Doob's guidance estimator $\hat{\mathbf{g}}_t^\lambda$ is characterized by the gradient-penalized regressor $\hat{h}_t^\lambda$ in (3.18), thus termed the regressor-guidance.

3.4 A summary of procedure. By a similar argument as Section 2.4, we have the exponential integrator for the controllable diffusion model:

$$(3.20) \quad \begin{aligned} d\hat{\mathbf{Z}}_t^\leftarrow &= (\hat{\mathbf{Z}}_t^\leftarrow + 2\hat{\mathbf{s}}(kh, \hat{\mathbf{Z}}_{kh}^\leftarrow) + 2\hat{\mathbf{g}}^\lambda(kh, \hat{\mathbf{Z}}_{kh}^\leftarrow)) dt + \sqrt{2} d\mathbf{B}_t, \quad t \in (kh, (k+1)h), \\ \hat{\mathbf{Z}}_0^\leftarrow &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \end{aligned}$$

where $0 \leq k \leq K-1$, the pre-trained base score estimator $\hat{\mathbf{s}}$ is defined as (2.4), and the Doob's guidance estimator $\hat{\mathbf{g}}^\lambda$ is defined as (3.19).

We apply post-processing to the generated particle $\hat{\mathbf{Z}}_{T-T_0}^\leftarrow$ to ensure numerical stability and facilitate the theoretical analysis presented in Theorem 4.5 (Lee et al., 2023a; Chen et al., 2023). First, we assume the target distribution q_0 is concentrated on a domain centered at the origin, such as a distribution with compact support (Assumption 1) or with light tails. Consequently, we introduce a truncation operator to the particles obtained from the controllable diffusion model (3.20). Second, because the controllable diffusion process is terminated at an early-stopping time T_0 , there exists a mean shift between the target distribution q_0 and the early-stopping distribution $q_{T_0} \approx \hat{q}_{T-T_0}$, as indicated by (2.2). To mitigate this drift, we employ a scaling operator. Specifically, for $R > 0$, we define a truncation operator $\mathcal{T}_R : \mathbf{z} \mapsto \mathbf{z} \mathbb{1}_{B(\mathbf{0}, R)}(\mathbf{z})$ and a scaling operator $\mathcal{M} : \mathbf{z} \mapsto \mu_{T_0}^{-1} \mathbf{z}$. The final processed particle is defined as

$$(3.21) \quad \mathcal{M} \circ \mathcal{T}_R(\hat{\mathbf{Z}}_{T-T_0}^\leftarrow) = \mu_{T_0}^{-1} \hat{\mathbf{Z}}_{T-T_0}^\leftarrow \mathbb{1}_{B(\mathbf{0}, R)}(\hat{\mathbf{Z}}_{T-T_0}^\leftarrow),$$

and we denote its density by $(\mathcal{M} \circ \mathcal{T}_R)_\# \hat{q}_{T-T_0}$.

A complete procedure is summarized in Algorithm 1.

Algorithm 1: Inference-time alignment via Doob's matching

Input: Base score estimator $\hat{\mathbf{s}}$, the weight function w , the regularization parameter λ , the step size h , and the number of steps K .
Output: Particle $\mathcal{M} \circ \mathcal{T}_R(\hat{\mathbf{Z}}_{T-T_0}^\leftarrow)$ follows the tilted distribution q_0 approximately.

- 1 **# Doob's matching**
- 2 Estimate Doob's h -function by $\hat{h}_t^\lambda$ via gradient penalized Doob's matching (3.18).
- 3 **# Controllable generation**
- 4 Generate the initial particle $\hat{\mathbf{Z}}_0^\leftarrow \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.
- 5 **for** $k = 0, \dots, K-1$ **do**
- 6 Evaluate the base score: $\hat{\mathbf{s}}_k \leftarrow \hat{\mathbf{s}}(kh, \hat{\mathbf{Z}}_{kh}^\leftarrow)$.
- 7 Evaluate the Doob's guidance: $\hat{\mathbf{g}}_k \leftarrow \nabla \log \hat{h}^\lambda(kh, \hat{\mathbf{Z}}_{kh}^\leftarrow)$.
- 8 Exponential integrator: $\hat{\mathbf{Z}}_{(k+1)h}^\leftarrow \sim \mathcal{N}(\exp(h)\hat{\mathbf{Z}}_{kh}^\leftarrow + 2\phi^2(h)(\hat{\mathbf{s}}_k + \hat{\mathbf{g}}_k), \phi^2(2h)\mathbf{I}_d)$.
- 9 **end**
- 10 **# Truncation and scaling**
- 11 $\mathcal{M} \circ \mathcal{T}_R(\hat{\mathbf{Z}}_{T-T_0}^\leftarrow) \leftarrow \mu_{T_0}^{-1} \hat{\mathbf{Z}}_{T-T_0}^\leftarrow \mathbb{1}_{B(\mathbf{0}, R)}(\hat{\mathbf{Z}}_{T-T_0}^\leftarrow)$.
- 12 **return** $\mathcal{M} \circ \mathcal{T}_R(\hat{\mathbf{Z}}_{T-T_0}^\leftarrow)$

4 Convergence Analysis

In this section, we establish a non-asymptotic convergence rate analysis for Doob's matching (3.18) and the induced controllable diffusion model (3.20) introduced in Section 3.

4.1 Assumptions. We begin by outlining the essential technical assumptions required for our theoretical results.

Assumption 1 (Bounded support). The support of the target distribution q_0 is a compact set contained within the hypercube $\{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_\infty \leq 1\}$.

Assumption 1 is a standard condition imposed on the data distribution (Lee et al., 2023a; Oko et al., 2023; Chang et al., 2025a; Beyler and Bach, 2025). This constraint is well-motivated by practical applications; for instance, image and video data consist of bounded pixel values, thereby satisfying this requirement.

Assumption 2 (Bounded weight function). The weight function w defined in (3.1) is bounded from above and bounded away from zero. Specifically, there exist constants $0 < \underline{B} < 1 < \bar{B} < \infty$ such that

$$\underline{B} \leq w(\mathbf{x}) \leq \bar{B}, \quad \text{for all } \mathbf{x} \in \text{supp}(q_0).$$

Assumption 2 implies that the base distribution p_0 and the tilted distribution q_0 satisfy mutual absolute continuity, ensuring that their supports coincide (i.e., $\text{supp}(p_0) = \text{supp}(q_0)$). This condition is crucial for establishing the regularity of Doob’s h -function. Furthermore, the ratio $\kappa := \bar{B}/\underline{B}$ serves as a condition number that characterizes the difficulty of the controllable diffusion task, as discussed in the context of posterior sampling by Purohit et al. (2025); Ding et al. (2024b); Chang et al. (2025b).

Assumption 3 (Base score estimation error). The base score estimator $\hat{\mathbf{s}}$ defined in (2.4) satisfies the following error bound:

$$\frac{1}{T} \sum_{k=0}^{K-1} h \mathbb{E}^{\mathbb{P}} \left[\|\hat{\mathbf{s}}(kh, \mathbf{X}_{kh}^{\leftarrow}) - \nabla \log p_{T-kh}(\mathbf{X}_{kh}^{\leftarrow})\|_2^2 \right] \leq \varepsilon_{\text{base}}^2.$$

Assumption 3 requires the L^2 -error of the base score estimator $\hat{\mathbf{s}}$ to be bounded with respect to the base path measure $\mathbb{P}$. In our setting, where numerous samples from the base distribution p_0 are available, estimators satisfying this bound can be obtained via implicit score matching (Hyvärinen, 2005), sliced score matching (Song et al., 2020), or denoising score matching (Vincent, 2011). While one can derive explicit bounds of base score matching as Tang and Yang (2024); Oko et al. (2023); Fu et al. (2024); Ding et al. (2024a); Yakovlev and Puchkin (2025a); Yakovlev et al. (2025) using non-parametric regression theory for deep neural networks (Bauer and Kohler, 2019; Schmidt-Hieber, 2020; Kohler and Langer, 2021; Jiao et al., 2023), we adopt this condition to maintain clarity of presentation, following the convention of Lee et al. (2023a); Chen et al. (2023); Beyler and Bach (2025); Kremling et al. (2025).

4.2 Regularity of Doob’s h -function. In this subsection, we establish the regularity properties of Doob’s h -function defined in (3.7).

Proposition 4.1. *Suppose Assumptions 1 and 2 hold. Then for all $t \in (0, T)$ and $\mathbf{x} \in \mathbb{R}^d$, the following bounds hold:*

- (i) $\underline{B} \leq h_t^*(\mathbf{x}) \leq \bar{B}$;
- (ii) $\max_{1 \leq k \leq d} |D_k h_t^*(\mathbf{x})| \leq 2\sigma_{T-t}^{-2} \bar{B}$; and
- (iii) $\max_{1 \leq k, \ell \leq d} |D_{k\ell}^2 h_t^*(\mathbf{x})| \leq 6\sigma_{T-t}^{-4} \bar{B}$,

where D_k and $D_{k\ell}^2$ denote the first-order and second-order partial derivatives with respect to the input coordinates, respectively.

The proof of Proposition 4.1 is deferred to Appendix C.1. It is worth noting that Proposition 4.1 relies solely on the boundedness of the weight function w , without requiring the existence or smoothness of its gradients. Nevertheless, we establish that Doob's h -function admits bounded derivatives. This result stems from the definition of Doob's h -function as a posterior expectation under a Gaussian likelihood; the inherent smoothness of the Gaussian kernel endows the posterior expectation with strong regularity properties.

4.3 Error bounds for the Doob's guidance estimator. We begin by introducing the concept of Rademacher complexity (Bartlett and Mendelson, 2002; Mohri et al., 2018), which is crucial for analyzing the generalization error.

Definition 1 (Rademacher complexity). Let $\mathcal{H}$ be a function class, and let $\mathbf{X}^{1:n} := (\mathbf{X}^1, \dots, \mathbf{X}^n)$ be a set of samples. The empirical Rademacher complexity of $\mathcal{H}$ with respect to $\mathbf{X}^{1:n}$ is defined as

$$\widehat{\mathfrak{R}}(\mathcal{H}|\mathbf{X}^{1:n}) := \mathbb{E} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(\mathbf{X}^i) \middle| \mathbf{X}^{1:n} \right],$$

where $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. Rademacher random variables. The Rademacher complexity of $\mathcal{H}$ is the expectation of empirical Rademacher complexity with respect to the distribution of $\mathbf{X}^{1:n}$ defined as

$$\mathfrak{R}_n(\mathcal{H}) := \mathbb{E}[\widehat{\mathfrak{R}}(\mathcal{H}|\mathbf{X}^{1:n})] = \mathbb{E} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(\mathbf{X}^i) \right].$$

For notational simplicity, we define the derivative classes and their associated Rademacher complexity as follows:

$$(4.1) \quad \mathfrak{R}_n(\nabla \mathcal{H}_t) := \sum_{k=1}^d \mathfrak{R}_n(D_k \mathcal{H}_t), \quad D_k \mathcal{H}_t := \{D_k h_t : h_t \in \mathcal{H}_t\}.$$

The following lemma provides an oracle inequality for the Doob's matching.

Lemma 4.2 (Oracle inequality). Suppose Assumptions 1 and 2 hold. Let $t \in (0, T)$ and let $\mathcal{H}_t$ be a convex hypothesis class. Let $\widehat{h}_t^\lambda$ be the gradient penalized empirical risk minimizer defined as (3.18), and let h_t^* be the Doob's h -function defined as (3.7). Then the following inequalities hold:

$$\begin{aligned} & \mathbb{E} \left[\|\widehat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2 \right] \\ & \lesssim \underbrace{\inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{H^1(p_{T-t})}^2}_{\text{approximation}} + \underbrace{\bar{B} \mathfrak{R}_n(\mathcal{H}_t) + \frac{\lambda \bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t)}_{\text{generalization}} + \underbrace{\frac{\lambda^2 d^2 \bar{B}^2}{\sigma_{T-t}^8}}_{\text{regularization}}, \\ & \mathbb{E} \left[\|\nabla \widehat{h}_t^\lambda - \nabla h_t^*\|_{L^2(p_{T-t})}^2 \right] \\ & \lesssim \underbrace{\frac{1}{\lambda} \inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{H^1(p_{T-t})}^2}_{\text{approximation}} + \underbrace{\frac{\bar{B}}{\lambda} \mathfrak{R}_n(\mathcal{H}_t) + \frac{\bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t)}_{\text{generalization}} + \underbrace{\frac{\lambda d^2 \bar{B}^2}{\sigma_{T-t}^8}}_{\text{regularization}}, \end{aligned}$$

where the notation $\lesssim$ hides absolute constants.

The proof of Lemma 4.2 is deferred to Appendix C.2. Both oracle inequalities for Doob's h -function and its gradient decompose the error into three components: approximation error, generalization error, and regularization error.

- (i) The approximation error is defined as the minimal H^1 -distance between functions in the hypothesis class $\mathcal{H}_t$ and the target function h_t^* , measuring the approximation capability of $\mathcal{H}_t$.
- (ii) The generalization error captures the error arising from finite-sample approximation, which vanishes as the number of samples approaches infinity.
- (iii) The regularization error is introduced by the gradient penalty in the objective functional, which causes the minimizer of the population risk to deviate from the exact Doob's h -function h_t^* (3.7).

Comparison with oracle inequality of vanilla regression. Lemma 4.2 is analogous to the oracle inequality found in regression problems. Let $\hat{h}_t$ be the estimator of the Doob's h -function obtained via vanilla regression (3.15). Informally, the following oracle inequality holds:

$$(4.2) \quad \mathbb{E} \left[\|\hat{h}_t - h_t^*\|_{L^2(p_{T-t})}^2 \right] \lesssim \underbrace{\inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{L^2(p_{T-t})}^2}_{\text{approximation}} + \underbrace{\bar{B}\mathfrak{R}_n(\mathcal{H}_t)}_{\text{generalization}}.$$

Comparing (4.2) with Lemma 4.2 reveals several crucial differences:

- (i) The approximation error in Lemma 4.2 is measured in the H^1 -norm, whereas in (4.2), it is measured in the L^2 -norm. This distinction is natural because we require the estimator to converge in the H^1 -norm; thus, simultaneous approximation of the function and its derivatives is essential. Simultaneous approximation using neural networks has been investigated in various contexts (Li et al., 2019; Gühring et al., 2020; Gühring and Raslan, 2021; Duan et al., 2022a,b; Lu et al., 2022b; Shen et al., 2022, 2024; Belomestny et al., 2023; Yakovlev and Puchkin, 2025b).
- (ii) In vanilla regression, the generalization error in (4.2) depends only on the complexity of the hypothesis class. In contrast, the generalization error bounds in Lemma 4.2 also depend on the complexity of the derivative classes (4.1). This occurs because the objective functional of the gradient penalized Doob's matching (3.16) includes the gradient norm term. Consequently, the error from finite-sample approximation is influenced not only by the complexity of the hypothesis class but also by that of the derivative class (4.1).
- (iii) The most significant difference lies in the regularization error. If we focus solely on the oracle inequality for $\hat{h}_t^\lambda$, letting λ go to zero reduces the expression to the vanilla regression oracle inequality (4.2). However, the bound for the gradient $\nabla \hat{h}_t^\lambda$ diverges as the regularization parameter λ approaches zero. This highlights the key advantage of our gradient penalized method: the gradient penalty is essential for guaranteeing simultaneous convergence of both the estimator value and its gradient. Additionally, there exists a trade-off with respect to λ in the oracle inequality for $\hat{h}_t^\lambda$: a larger λ leads to larger regularization error, while reduces the approximation and generalization errors.

Given the oracle inequality for a general hypothesis class $\mathcal{H}_t$, we consider the specific case that $\mathcal{H}_t$ is chosen as a neural network class, with the aim of deriving non-asymptotic convergence rates. We begin by formally defining the neural network class.

Definition 2 (Neural network class). A function implemented by a neural network $h : \mathbb{R}^{N_0} \rightarrow \mathbb{R}^{N_{L+1}}$ is defined by

$$h(\mathbf{x}) = T_L(\varrho(T_{L-1}(\cdots \varrho(T_0(\mathbf{x})) \cdots))),$$

where the activation function ϱ is applied component-wise and $T_\ell(\mathbf{x}) := \mathbf{A}_\ell \mathbf{x} + \mathbf{b}_\ell$ is an affine transformation with $\mathbf{A}_\ell \in \mathbb{R}^{N_{\ell+1} \times N_\ell}$ and $\mathbf{b}_\ell \in \mathbb{R}^{N_\ell}$ for $\ell = 0, \dots, L$. In this paper, we consider the case where $N_0 = d + 1$ and $N_{L+1} = 1$. The number L is called the depth of neural networks. Additionally, $S := \sum_{\ell=0}^L (\|\mathbf{A}_\ell\|_0 + \|\mathbf{b}_\ell\|_0)$ represents the total number of non-zero weights within the neural network. We denote by $N(L, S)$ the set of neural networks with depth at most L and the number of non-zero weights at most S .

The following theorem establishes the convergence rate of the estimated Doob's guidance given in (3.19).

Theorem 4.3 (Convergence rate of Doob's guidance). *Suppose Assumptions 1 and 2 hold. Let $t \in (0, T)$. Set the hypothesis class $\mathcal{H}_t$ as*

$$(4.3) \quad \mathcal{H}_t := \left\{ h_t \in \text{conv}(N(L, S)) : \begin{array}{l} \sup_{\mathbf{x} \in \mathbb{R}^d} h_t(\mathbf{x}) \leq \bar{B}, \quad \inf_{\mathbf{x} \in \mathbb{R}^d} h_t(\mathbf{x}) \geq \underline{B}, \\ \max_{1 \leq k \leq d} \sup_{\mathbf{x} \in \mathbb{R}^d} |D_k h_t(\mathbf{x})| \leq 2\sigma_{T-t}^{-2} \bar{B}, \end{array} \right\}.$$

where $L = \mathcal{O}(\log n)$ and $S = \mathcal{O}(n^{\frac{d}{d+4}})$. Let $\hat{h}_t^\lambda$ be the gradient penalized empirical risk minimizer defined as (3.18), and let h_t^* be the Doob's h -function defined as (3.7). Then the following inequalities hold:

$$\mathbb{E} \left[\|\nabla \log \hat{h}_t^\lambda - \nabla \log h_t^*\|_{L^2(p_{T-t})}^2 \right] \leq C \sigma_{T-t}^{-8} n^{-\frac{1}{d+4}},$$

provided that the regularization parameter λ is set as $\lambda = \mathcal{O}(n^{-\frac{1}{d+4}})$, where C is a constant depending only on d , $\bar{B}$, and $\underline{B}$.

The proof of Theorem 4.3 is deferred to Appendix C.4. This theorem demonstrates that the L^2 -error of the Doob's guidance estimator (3.19) converges to the exact Doob's guidance in (3.11) as the sample size increases, provided that the size of the neural network is appropriately chosen. However, since the prefactor σ_{T-t}^{-8} diverges as t approaches the terminal time T , early stopping in controllable diffusion models (3.20) is required to ensure the validity of the Doob's guidance estimator.

4.4 Error bounds for the controllable diffusion models. In this subsection, we establish a non-asymptotic convergence rate for the controllable diffusion models (3.20). We begin by presenting an error decomposition for the KL-divergence between the early-stopping distribution q_{T_0} and the distribution of $\hat{\mathbf{Z}}_{T-T_0}^\leftarrow$.

Lemma 4.4 (Error decomposition). *Suppose Assumptions 1, 2, and 3 hold. Let $\hat{q}_{T-T_0}$ be the marginal density of $\hat{\mathbf{Z}}_{T-T_0}^\leftarrow$ defined in (3.20). Then it follows that*

$$\begin{aligned} \text{KL}(q_{T_0} \parallel \hat{q}_{T-T_0}) &\lesssim \underbrace{\frac{\bar{B}}{B} \sum_{k=0}^{K-1} h \mathbb{E}^\mathbb{P} \left[\|\nabla \log \hat{h}_{kh}(\mathbf{X}_{kh}^\leftarrow) - \nabla \log h_{kh}^*(\mathbf{X}_{kh}^\leftarrow)\|_2^2 \right]}_{(I)} \\ &\quad + \underbrace{\frac{\bar{B}}{B} T \varepsilon_{\text{base}}^2}_{(II)} + \underbrace{d \exp(-T)}_{(III)} + \underbrace{\frac{d^2 T^2}{\sigma_{T_0}^4 K}}_{(IV)}, \end{aligned}$$

where the notation $\lesssim$ hides absolute constants.

The proof of Lemma 4.4 is deferred to Appendix D.1. Lemma 4.4 decomposes the KL-divergence between the early-stopping distribution q_{T_0} and the distribution of $\hat{\mathbf{Z}}_{T-T_0}^\leftarrow$ into four components: (I) Doob's guidance error, (II) the base score error, (III) the initialization error, and (IV) the discretization error. Specifically, Doob's guidance error represents the average error of Doob's h -guidance estimator at each time point, which has been investigated in Theorem 4.3; the base score error is the average error of the base score estimator at each time point, which is discussed in Assumption 3; the initialization error arises from replacing the initial distribution $q_T = p_T$ with a Gaussian distribution in (3.20); and the discretization error is induced by the exponential integrator.

While Lemma 4.4 characterizes the error between the estimated distribution $\hat{q}_{T-T_0}$ and the early-stopping distribution q_{T_0} , our primary interest lies in the discrepancy between $\hat{q}_{T-T_0}$ and the target tilted distribution q_0 defined in (3.1). Since the KL-divergence does not satisfy the triangular inequality, we instead propose an error bound in 2-Wasserstein distance. The following theorem establishes the 2-Wasserstein distance between the scaled and truncated distribution $(\mathcal{M} \circ \mathcal{T}_R)_\# \hat{q}_{T-T_0}$ defined in (3.21) and the target tilted distribution q_0 .

Theorem 4.5 (Convergence rate of controllable diffusion models). *Suppose Assumptions 1, 2, and 3 hold. Let $\varepsilon \in (0, 1)$. Set the hypothesis classes $\{\mathcal{H}_{T-kh}\}_{k=0}^{K-1}$ as (4.3) with the same depth L and number of non-zero parameters S as Theorem 4.3. Let $\hat{q}_{T-T_0}$ be the marginal density of $\hat{\mathbf{Z}}_{T-T_0}^\leftarrow$ defined in (3.20), and let $(\mathcal{M} \circ \mathcal{T}_R)_\# \hat{q}_{T-T_0}$ defined as (3.21). Then it follows that*

$$(4.4) \quad \mathbb{E} \left[\mathcal{W}_2^2(q_0, (\mathcal{M} \circ \mathcal{T}_R)_\# \hat{q}_{T-T_0}) \right] \leq C \left\{ \sigma_{T_0}^2 + \varepsilon + \frac{\varepsilon^3}{\sigma_{T_0}^4} \log \left(\frac{1}{\varepsilon} \right) \right\},$$

provided that the truncation radius R , the step size h , the number of steps K , the error of base score $\varepsilon_{\text{base}}$, and the number of samples n for Doob's matching are set, respectively, as

$$\begin{aligned} R &= \Theta \left(\log^{\frac{1}{2}} \left(\frac{1}{\varepsilon} \right) \right), \quad h = \Theta \left(\varepsilon^4 \log^{-1} \left(\frac{1}{\varepsilon} \right) \right), \quad K = \Theta \left(\frac{1}{\varepsilon^4} \log^2 \left(\frac{1}{\varepsilon} \right) \right), \\ \varepsilon_{\text{base}}^2 &= \Theta \left(\varepsilon^2 \log^{-1} \left(\frac{1}{\varepsilon} \right) \right), \quad n = \Theta \left(\frac{1}{\varepsilon^{6(d+4)}} \log^{d+4} \left(\frac{1}{\varepsilon} \right) \right), \end{aligned}$$

Further, letting $T_0 = \Theta(\varepsilon)$, it holds that

$$(4.5) \quad \mathbb{E} \left[\mathcal{W}_2^2(q_0, (\mathcal{M} \circ \mathcal{T}_R)_\# \hat{q}_{T-T_0}) \right] \leq C \varepsilon \log \left(\frac{1}{\varepsilon} \right).$$

Here C is a constant depending only on d , $\bar{B}$, and B .

The proof of Theorem 4.5 is deferred to Appendix D.2. In this theorem, (4.4) illustrates a trade-off with respect to the early-stopping time T_0 . Specifically, as T_0 approaches zero, the early-stopping error $\mathcal{O}(\sigma_{T_0}^2)$ vanishes, while the second term blows up. By balancing these competing terms, we obtain the optimal convergence rate in (4.5) and provide theoretical guidance for the choice of the early-stopping time. This theorem provides theoretical guidance for the selection of hyper-parameters, including the truncation radius R in (3.21), the step size h , the number of steps K in (3.20), the error of base score $\varepsilon_{\text{base}}$ in Assumption 3, and the number of samples n for Doob’s matching in (3.17).

5 Conclusions

In this work, we proposed Doob’s matching, a principled inference-time alignment framework for diffusion models grounded in the theory of Doob’s h -transform. Our approach reformulates guidance as the gradient of the logarithm of an underlying Doob’s h -function, providing a mathematically consistent mechanism for tilting a pre-trained diffusion model toward a target distribution without retraining the base score network. By leveraging gradient-penalized regression, Doob’s matching simultaneously estimates both the h -function and its gradient, thereby providing a consistent estimator for Doob’s guidance.

From a theoretical perspective, we established non-asymptotic convergence rates for the proposed guidance estimator, showing that the estimated Doob’s guidance converges to the true guidance under suitable choices of the hypothesis class and regularization parameter. Building on this result, we further derived non-asymptotic convergence guarantees for the induced controllable diffusion process, demonstrating that the generated distribution converges to the target distribution in the 2-Wasserstein distance. These results provide an end-to-end theoretical guarantees for inference-time aligned diffusion models that explicitly account for guidance estimation error, base score estimation error, initialization bias, and discretization error.

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Outline of Appendices

A Derivations in Section 2

Lemma A.1 (Exponential integrator). *Suppose $\mathbf{X}_t$ satisfies a linear SDE:*

$$d\mathbf{X}_t = (\mathbf{X}_t + \mathbf{b}(s, \mathbf{X}_s)) dt + \sqrt{2} d\mathbf{B}_t, \quad t \geq s,$$

Then it follows that

$$\mathbf{X}_t = \exp(t - s)\mathbf{X}_s + \phi^2(t - s)\mathbf{b}(s, \mathbf{X}_s) + \phi(2(t - s))\boldsymbol{\xi},$$

where $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ and $\phi(z) = \sqrt{\exp(z) - 1}$.

B Derivations in Section 3

Lemma B.1. *The Doob’s h -transform $h^*(t, \mathbf{X}_t^\leftarrow)$ defined as (3.7) is a martingale, and satisfies the following SDE:*

$$dh^*(t, \mathbf{X}_t^\leftarrow) = \nabla h^*(t, \mathbf{X}_t^\leftarrow)^\top \sqrt{2} d\mathbf{B}_t.$$

Proof of Lemma B.1. This proof is divided into two parts.

Part 1. Martingale. Due to the Markov property of the diffusion process $\mathbf{X}_t^\leftarrow$ (Øksendal, 2003, Theorem 7.1.2), using (3.7) implies

$$M_t := h^*(t, \mathbf{X}_t^\leftarrow) = \mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow) | \mathcal{F}_t].$$

It is apparent that M_t is $\mathcal{F}_t$ -measurable for each $t \in (0, T)$, thus M_t is adapted to $\mathbb{F}$. Then we show that M_t is integrable under $\mathbb{P}$. Indeed,

$$\mathbb{E}^\mathbb{P}[|M_t|] \leq \mathbb{E}^\mathbb{P}[\mathbb{E}^\mathbb{P}[|w(\mathbf{X}_T^\leftarrow)| | \mathcal{F}_t]] = \mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow)] = Z < \infty,$$

where the first inequality holds from Jensen's inequality, and the first equality used the law of total expectation and the fact that $w(\mathbf{X}_T^\leftarrow)$. We next show the martingale property. For each $0 \leq s \leq t \leq T$,

$$\mathbb{E}^\mathbb{P}[M_t | \mathcal{F}_s] = \mathbb{E}^\mathbb{P}[\mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow) | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow) | \mathcal{F}_s] = M_s,$$

where the second equality involves the tower property of conditional expectation, and the fact that $\mathcal{F}_s \subseteq \mathcal{F}_t$. Therefore, M_t is a martingale.

Part 2. Stochastic dynamics. Applying Itô's formula to $h^*(t, \mathbf{X}_t^\leftarrow)$ yields

$$\begin{aligned} dh^*(t, \mathbf{X}_t^\leftarrow) &= \partial_t h^*(t, \mathbf{X}_t^\leftarrow) dt + \nabla h^*(t, \mathbf{X}_t^\leftarrow)^\top d\mathbf{X}_t^\leftarrow + \frac{1}{2} (d\mathbf{X}_t^\leftarrow)^\top \nabla^2 h^*(t, \mathbf{X}_t^\leftarrow) d\mathbf{X}_t^\leftarrow \\ &= \nabla h^*(t, \mathbf{X}_t^\leftarrow)^\top \sqrt{2} d\mathbf{B}_t, \end{aligned}$$

where the last equality holds from the fact that martingale has zero drift. This completes the proof. $\square$

Proposition 3.1. *The Radon-Nikodym derivative process L_t , as defined in (3.5), admits the following representation:*

$$L_t = \mathbb{E}^\mathbb{P} \left[\frac{d\mathbf{Q}}{d\mathbb{P}} \middle| \mathcal{F}_t \right] = \frac{h^*(t, \mathbf{X}_t^\leftarrow)}{\mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow)]},$$

where $h^* : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$, referred to as Doob's h -function, is defined as the conditional expectation of the terminal weight:

$$h^*(t, \mathbf{x}) := \mathbb{E}^\mathbb{P}[w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow = \mathbf{x}].$$

Furthermore, the log-likelihood ratio satisfies the following SDE:

$$d(\log L_t) = \nabla \log h^*(t, \mathbf{X}_t^\leftarrow)^\top \sqrt{2} d\mathbf{B}_t - \|\nabla \log h^*(t, \mathbf{X}_t^\leftarrow)\|_2^2 dt.$$

Proof of Proposition 3.1. The proof is divided into three parts.

Part 1. The equivalent definition of L_t . In this part, we aim to show the conditional expectation in Proposition 3.1 is identical to the likelihood L_t defined as (3.5). Define

$$\bar{L}_t = \mathbb{E}^\mathbb{P} \left[\frac{d\mathbf{Q}}{d\mathbb{P}} \middle| \mathcal{F}_t \right].$$

For each event $A \in \mathcal{F}_t \subseteq \mathcal{F}_T$, we have

$$\mathbf{Q}(A) = \int_A \frac{d\mathbf{Q}}{d\mathbb{P}} d\mathbb{P} = \int_A \mathbb{E} \left[\frac{d\mathbf{Q}}{d\mathbb{P}} \middle| \mathcal{F}_t \right] d\mathbb{P} = \int_A \bar{L}_t d\mathbb{P},$$

where the second equality is due to $A \in \mathcal{F}_t$. This means $\bar{L}_t$ acts as the density for the measure $\mathbb{Q}$ with respect to $\mathbb{P}$ when restricting to the σ -algebra $\mathcal{F}_t$. Thus $L_t \equiv \bar{L}_t$ for each $t \in (0, T)$.

Part 2. The expression of Doob's h -transform. It is straightforward that

$$L_t = \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \middle| \mathcal{F}_t \right] = \frac{\mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow}) | \mathcal{F}_t]}{\mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow})]} = \frac{\mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow}) | \mathbf{X}_t^{\leftarrow}]}{\mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow})]} = \frac{h^*(t, \mathbf{X}_t^{\leftarrow})}{\mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow})]},$$

where the second equality follows from (3.6), the third equality involves the Markovity of the diffusion process $\mathbf{X}_t^{\leftarrow}$, and the last equality is due to the definition of the h -function (3.7).

Part 3. The stochastic dynamics of log-likelihood. We can now derive the dynamics of the likelihood L_t . Letting $Z = \mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow})]$, we have:

$$\begin{aligned} dL_t &= \frac{1}{Z} dh^*(t, \mathbf{X}_t^{\leftarrow}) \\ &= \frac{1}{Z} (\nabla h^*(t, \mathbf{X}_t^{\leftarrow}))^\top \sqrt{2} d\mathbf{B}_t \\ &= \frac{h^*(t, \mathbf{X}_t^{\leftarrow})}{Z} \frac{\nabla h^*(t, \mathbf{X}_t^{\leftarrow})^\top}{h^*(t, \mathbf{X}_t^{\leftarrow})} \sqrt{2} d\mathbf{B}_t \\ (B.1) \quad &= L_t (\nabla \log h^*(t, \mathbf{X}_t^{\leftarrow}))^\top \sqrt{2} d\mathbf{B}_t, \end{aligned}$$

where the second equality is due to Lemma B.1. Using Itô's formula for log-likelihood yields

$$\begin{aligned} d(\log L_t) &= \frac{dL_t}{L_t} - \frac{1}{2} \left\langle \frac{dL_t}{L_t}, \frac{dL_t}{L_t} \right\rangle \\ &= \sqrt{2} \nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})^\top d\mathbf{B}_t - \frac{1}{2} \|\sqrt{2} \nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})\|_2^2 dt \\ &= \sqrt{2} \nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})^\top d\mathbf{B}_t - \|\nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})\|_2^2 dt, \end{aligned}$$

where the second equality is due to (B.1). This completes the proof. $\square$

Proposition 3.2. *Let the base process $\mathbf{X}_t^{\leftarrow}$ satisfy the SDE (2.3) under the path measure $\mathbb{P}$, and let $\mathbb{Q}$ be the target path measure defined by (3.6). Assume that the Novikov condition holds:*

$$\mathbb{E}^{\mathbb{P}} \left[\exp \left(\int_0^T \|\nabla \log h^*(s, \mathbf{X}_s^{\leftarrow})\|_2^2 ds \right) \right] < \infty.$$

Define a process $(\tilde{\mathbf{B}}_t)_{1 \leq t \leq T}$ by

$$d\tilde{\mathbf{B}}_t = d\mathbf{B}_t - \sqrt{2} \nabla \log h^*(t, \mathbf{X}_t^{\leftarrow}) dt,$$

where $\mathbf{B}_t$ is a Brownian motion under $\mathbb{P}$. Then $\tilde{\mathbf{B}}_t$ is a standard Brownian motion under $\mathbb{Q}$. Further, under the path measure $\mathbb{Q}$, the base process $\mathbf{X}_t^{\leftarrow}$ in (2.3) evolves according to:

$$d\mathbf{X}_t^{\leftarrow} = (\mathbf{X}_t^{\leftarrow} + 2\nabla \log p_{T-t}(\mathbf{X}_t^{\leftarrow}) + 2\nabla \log h^*(t, \mathbf{X}_t^{\leftarrow})) dt + \sqrt{2} d\tilde{\mathbf{B}}_t.$$

Proof of Proposition 3.2. The derivation proceeds in two steps.

Step 1. The Brownian motion under the target path measure $\mathbb{Q}$. Using Proposition 3.1 and noting that $L_0 \equiv 1$ as (3.6), we have

$$L_t = \exp \left(\int_0^t \nabla \log h^*(s, \mathbf{X}_s^{\leftarrow})^\top \sqrt{2} d\mathbf{B}_s - \|\nabla \log h^*(s, \mathbf{X}_s^{\leftarrow})\|_2^2 ds \right).$$

According to Karatzas and Shreve (1998, Corollary 5.13), the Novikov condition (3.8) implies that L_t is a martingale. Then applying Girsanov's theorem (Øksendal, 2003, Theorem 8.6.6) yields that $\tilde{\mathbf{B}}_t$ is a Brownian motion under $\mathbf{Q}$.

Step 2. Dynamics under the target path measure. Recall that under the base path measure $\mathbb{P}$, the process evolves as

$$d\mathbf{X}_t^\leftarrow = (\mathbf{X}_t^\leftarrow + 2\nabla \log p_{T-t}(\mathbf{X}_t^\leftarrow)) dt + \sqrt{2} d\mathbf{B}_t,$$

where $\mathbf{B}_t$ is a $\mathbb{P}$ -Brownian motion. Substituting the relationship (3.9) into this SDE implies

$$\begin{aligned} d\mathbf{X}_t^\leftarrow &= (\mathbf{X}_t^\leftarrow + 2\nabla \log p_{T-t}(\mathbf{X}_t^\leftarrow)) dt + \sqrt{2}(d\tilde{\mathbf{B}}_t + \sqrt{2}\nabla \log h^*(t, \mathbf{X}_t^\leftarrow) dt) \\ &= (\mathbf{X}_t^\leftarrow + 2\nabla \log p_{T-t}(\mathbf{X}_t^\leftarrow) + 2\nabla \log h^*(t, \mathbf{X}_t^\leftarrow)) dt + \sqrt{2} d\tilde{\mathbf{B}}_t, \end{aligned}$$

which is the dynamics of $\mathbf{X}_t^\leftarrow$ under the target path measure $\mathbf{Q}$. This completes the proof. $\square$

Proposition 3.4. *For every $t \in (0, T)$, the Doob's h -function h_t^* in (3.7) minimizes the implicit Doob's matching loss (3.13). Further,*

$$\mathcal{J}_t(h_t) = \mathbb{E}^\mathbb{P} [\|h_t(\mathbf{X}_t^\leftarrow) - h_t^*(\mathbf{X}_t^\leftarrow)\|_2^2] + V_t^2,$$

where $V_t^2 := \mathbb{E}^\mathbb{P} [\text{Var}(w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow)]$ is a constant independent of h_t .

Proof of Proposition 3.4. By a direct calculation, we have

$$\begin{aligned} \mathcal{J}_t(h_t) &= \mathbb{E}^\mathbb{P} [\|h_t(\mathbf{X}_t^\leftarrow) - w(\mathbf{X}_T^\leftarrow)\|_2^2] \\ &= \mathbb{E}^\mathbb{P} [\|h_t(\mathbf{X}_t^\leftarrow) - h_t^*(\mathbf{X}_t^\leftarrow) + h_t^*(\mathbf{X}_t^\leftarrow) - w(\mathbf{X}_T^\leftarrow)\|_2^2] \\ (B.2) \quad &= \mathbb{E}^\mathbb{P} [\|h_t(\mathbf{X}_t^\leftarrow) - h_t^*(\mathbf{X}_t^\leftarrow)\|_2^2] + \mathbb{E}^\mathbb{P} [\|h_t^*(\mathbf{X}_t^\leftarrow) - w(\mathbf{X}_T^\leftarrow)\|_2^2] \\ &\quad + 2\mathbb{E}^\mathbb{P} [\langle h_t(\mathbf{X}_t^\leftarrow) - h_t^*(\mathbf{X}_t^\leftarrow), h_t^*(\mathbf{X}_t^\leftarrow) - w(\mathbf{X}_T^\leftarrow) \rangle], \end{aligned}$$

where h_t^* is defined as (3.7). For the second term in (B.2), we have

$$\begin{aligned} &\mathbb{E}^\mathbb{P} [\|h_t^*(\mathbf{X}_t^\leftarrow) - w(\mathbf{X}_T^\leftarrow)\|_2^2] \\ &= \mathbb{E}^\mathbb{P} [\|\mathbb{E}^\mathbb{P} [w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow] - w(\mathbf{X}_T^\leftarrow)\|_2^2] \\ &= \mathbb{E}^\mathbb{P} \mathbb{E}^\mathbb{P} [\|\mathbb{E}^\mathbb{P} [w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow] - w(\mathbf{X}_T^\leftarrow)\|_2^2 | \mathbf{X}_t^\leftarrow] \\ (B.3) \quad &= \mathbb{E}^\mathbb{P} [\text{Var}(w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow)], \end{aligned}$$

where the first equality holds from the definition of h_t^* (3.7), and the second equality is due to the law of the total expectation. For the third term in (B.2), we find

$$\begin{aligned} &\mathbb{E}^\mathbb{P} [\langle h_t(\mathbf{X}_t^\leftarrow) - h_t^*(\mathbf{X}_t^\leftarrow), h_t^*(\mathbf{X}_t^\leftarrow) - w(\mathbf{X}_T^\leftarrow) \rangle] \\ &= \mathbb{E}^\mathbb{P} [\langle h_t(\mathbf{X}_t^\leftarrow) - \mathbb{E}^\mathbb{P} [w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow], \mathbb{E}^\mathbb{P} [w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow] - w(\mathbf{X}_T^\leftarrow) \rangle] \\ &= \mathbb{E}^\mathbb{P} [\langle h_t(\mathbf{X}_t^\leftarrow) - \mathbb{E}^\mathbb{P} [w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow], \mathbb{E}^\mathbb{P} [w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow] - \mathbb{E}^\mathbb{P} [w(\mathbf{X}_T^\leftarrow) | \mathbf{X}_t^\leftarrow] \rangle] \\ (B.4) \quad &= 0, \end{aligned}$$

where the first equality holds from the definition of h_t^* (3.7), and the second equality is due to the law of the total expectation. Substituting (B.3) and (B.4) into (B.2) completes the proof. $\square$

C Error Bounds for the Doob's Guidance Estimation

C.1 The regularity of the Doob's h -function.

Lemma C.1. *Suppose Assumption 2 holds. Then for all $t \in (0, T)$,*

$$\underline{B} \leq h_t^*(\mathbf{x}) \leq \bar{B}, \quad \mathbf{x} \in \mathbb{R}^d.$$

Proof of Lemma C.1. A direct conclusion of Assumption 2 and the definition of Doob's h -function h_t^* (3.7). $\square$

Lemma C.2 (Tweedie's formula). *Let $t \in (0, T)$, and let $\mathbf{X}_t$ be defined as (2.1). Then*

$$\nabla \log p_t(\mathbf{x}) + \frac{\mathbf{x}}{\sigma_t^2} = \frac{\mu_t}{\sigma_t^2} \mathbb{E}[\mathbf{X}_0 | \mathbf{X}_t = \mathbf{x}], \quad \mathbf{x} \in \mathbb{R}^d.$$

Proof of Lemma C.2. It is straightforward that

$$\begin{aligned} \nabla \log p_t(\mathbf{x}) &= \frac{\nabla p_t(\mathbf{x})}{p_t(\mathbf{x})} \\ &= \frac{1}{p_t(\mathbf{x})} \int \nabla_{\mathbf{x}} \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0 \\ &= -\frac{1}{p_t(\mathbf{x})} \int \left(\frac{\mathbf{x} - \mu_t \mathbf{x}_0}{\sigma_t^2} \right) \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0 \\ &= -\frac{1}{p_t(\mathbf{x})} \frac{\mathbf{x}}{\sigma_t^2} \int \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0 \\ &\quad + \frac{\mu_t}{\sigma_t^2} \int \mathbf{x}_0 \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) \frac{p_0(\mathbf{x}_0)}{p_t(\mathbf{x})} d\mathbf{x}_0 \\ &= -\frac{\mathbf{x}}{\sigma_t^2} + \frac{\mu_t}{\sigma_t^2} \mathbb{E}[\mathbf{X}_0 | \mathbf{X}_t = \mathbf{x}], \end{aligned}$$

where the second equality is due to (2.2), and last equality invokes the Bayes' rule. This completes the proof. $\square$

Lemma C.3. *Let $g : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an integrable function. Let $t \in (0, T)$, and let $\mathbf{X}_t$ be defined as (2.1). Then for each $\mathbf{x} \in \mathbb{R}^d$,*

$$\nabla_{\mathbf{x}} \mathbb{E}[g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x}] = \frac{\mu_t}{\sigma_t^2} \text{Cov}(\mathbf{X}_0, g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x}),$$

where the k -th entry of $\text{Cov}(\mathbf{X}_0, g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x})$ is defined as $\text{Cov}(X_{0,k}, g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x})$ with $\mathbf{X}_0 = (X_{0,1}, \dots, X_{0,d})$.

Proof of Lemma C.3. According to Bayes' rule, we have

$$\mathbb{E}[g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x}] = \frac{1}{p_t(\mathbf{x})} \int g(\mathbf{x}_0) \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0.$$

Taking gradient with respect to $\mathbf{x}$ on both sides of the equality yields

$$\begin{aligned}
 \nabla_{\mathbf{x}} \mathbb{E}[g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x}] &= \frac{1}{p_t(\mathbf{x})} \int g(\mathbf{x}_0) \nabla_{\mathbf{x}} \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 &\quad - \frac{\nabla p_t(\mathbf{x})}{p_t^2(\mathbf{x})} \int g(\mathbf{x}_0) \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 &= -\frac{1}{p_t(\mathbf{x})} \int g(\mathbf{x}_0) \left(\frac{\mathbf{x} - \mu_t \mathbf{x}_0}{\sigma_t^2} \right) \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 &\quad - \frac{\nabla \log p_t(\mathbf{x})}{p_t(\mathbf{x})} \int g(\mathbf{x}_0) \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 &= -\left(\nabla \log p_t(\mathbf{x}) + \frac{\mathbf{x}}{\sigma_t^2} \right) \int g(\mathbf{x}_0) \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) \frac{p_0(\mathbf{x}_0)}{p_t(\mathbf{x})} d\mathbf{x}_0 \\
 &\quad + \frac{\mu_t}{\sigma_t^2} \int \mathbf{x}_0 g(\mathbf{x}_0) \varphi_d(\mathbf{x}; \mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d) \frac{p_0(\mathbf{x}_0)}{p_t(\mathbf{x})} d\mathbf{x}_0 \\
 &= \frac{\mu_t}{\sigma_t^2} (-\mathbb{E}[\mathbf{X}_0 | \mathbf{X}_t = \mathbf{x}] \mathbb{E}[g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x}] + \mathbb{E}[\mathbf{X}_0 g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x}]) \\
 &= \frac{\mu_t}{\sigma_t^2} \text{Cov}(\mathbf{X}_0, g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x}),
 \end{aligned}$$

where the fourth equality follows from Lemma C.2 and the Bayes' rule. Here the k -th entry of the conditional covariance $\text{Cov}(\mathbf{X}_0, g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x})$ is defined as $\text{Cov}(X_{0,k}, g(\mathbf{X}_0) | \mathbf{X}_t = \mathbf{x})$, where $\mathbf{X}_0 = (X_{0,1}, \dots, X_{0,d})$. This completes the proof. $\square$

Lemma C.4. Suppose Assumptions 1 and 2 hold. Then for each $\mathbf{x} \in \mathbb{R}^d$ and $t > 0$,

$$\max_{1 \leq k \leq d} |D_k h_t^*(\mathbf{x})| \leq \frac{2\bar{B}}{\sigma_{T-t}^2},$$

where D_k denote the differential operator with respect to the k -th entry of $\mathbf{x}$.

Proof of Lemma C.4. According to the definition of Doob's h -function h_t^* (3.7) and the property of the time-reversal process (2.3), we have

$$h^*(t, \mathbf{x}) := \mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow}) | \mathbf{X}_t^{\leftarrow} = \mathbf{x}] = \mathbb{E}[w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}],$$

where the second expectation is with respect to the path measure of the forward process (2.1). Then it follows from Lemma C.3 that

$$(C.1) \quad \nabla h_t^*(\mathbf{x}) = \frac{\mu_{T-t}}{\sigma_{T-t}^2} \text{Cov}(\mathbf{X}_0, w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}).$$

Then it follows from Assumptions 1 and 2 that for each $\mathbf{x} \in \mathbb{R}^d$,

$$(C.2) \quad \|\text{Cov}(\mathbf{X}_0, w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x})\|_{\infty} = \max_{1 \leq k \leq d} \text{Cov}(X_{0,k}, w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}) \leq 2\bar{B}.$$

Substituting (C.2) into (C.1) yields

$$\|\nabla h_t^*(\mathbf{x})\|_{\infty} = \frac{\mu_t}{\sigma_t^2} \|\text{Cov}(\mathbf{X}_0, w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x})\|_{\infty} \leq \frac{2\bar{B}}{\sigma_{T-t}^2},$$

where we used the fact that $\mu_t = \exp(-t) < 1$. This completes the proof. $\square$

Lemma C.5. Suppose Assumptions 1 and 2 hold. Then for each $\mathbf{x} \in \mathbb{R}^d$ and $t > 0$,

$$|D_{k\ell}^2 h_t^*(\mathbf{x})| \leq \frac{6\bar{B}}{\sigma_{T-t}^4},$$

where $D_{k\ell}^2$ denote the second-order differential operator with respect to k -th and ℓ -th entry.

Proof of Lemma C.5. Taking derivative with respect to the ℓ -th entry of $\mathbf{x}$ on both sides of (C.1) implies

$$(C.3) \quad D_{k\ell}^2 h_t^*(\mathbf{x}) = \frac{\mu_{T-t}}{\sigma_{T-t}^2} D_\ell \text{Cov}(X_{0,k}, w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}).$$

It remains to estimate the derivative of the conditional covariance. Indeed,

$$\begin{aligned} & D_\ell \text{Cov}(X_{0,k}, w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}) \\ &= D_\ell \mathbb{E}[X_{0,k} | \mathbf{X}_{T-t} = \mathbf{x}] \mathbb{E}[w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}] \\ &\quad + \mathbb{E}[X_{0,k} | \mathbf{X}_{T-t} = \mathbf{x}] D_\ell \mathbb{E}[w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}] \\ &\quad - D_\ell \mathbb{E}[X_{0,k} w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}] \\ &= \frac{\mu_{T-t}}{\sigma_{T-t}^2} \text{Cov}(X_{0,\ell}, X_{0,k} | \mathbf{X}_t = \mathbf{x}) \mathbb{E}[w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x}] \\ &\quad + \mathbb{E}[X_{0,k} | \mathbf{X}_{T-t} = \mathbf{x}] \frac{\mu_{T-t}}{\sigma_{T-t}^2} \text{Cov}(w(\mathbf{X}_0), X_{0,\ell} | \mathbf{X}_{T-t} = \mathbf{x}) \\ &\quad - \frac{\mu_{T-t}}{\sigma_{T-t}^2} \text{Cov}(X_{0,k} w(\mathbf{X}_0), X_{0,\ell} | \mathbf{X}_{T-t} = \mathbf{x}), \end{aligned}$$

where the last equality holds from Lemma C.3. Consequently, for each $\mathbf{x} \in \mathbb{R}^d$ and $t > 0$,

$$(C.4) \quad |D_\ell \text{Cov}(X_{0,k}, w(\mathbf{X}_0) | \mathbf{X}_{T-t} = \mathbf{x})| \leq \frac{6\mu_{T-t}\bar{B}}{\sigma_{T-t}^2},$$

where we used Assumptions 1 and 2. Substituting (C.4) into (C.3) completes the proof. $\square$

C.2 Oracle inequality of the Doob's h -function matching.

Lemma 4.2. Suppose Assumptions 1 and 2 hold. Let $t \in (0, T)$ and let $\mathcal{H}_t$ be a convex hypothesis class. Let $\hat{h}_t^\lambda$ be the gradient penalized empirical risk minimizer defined as (3.18), and let h_t^* be the Doob's h -function defined as (3.7). Then the following inequalities hold:

$$\begin{aligned} & \mathbb{E} \left[\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2 \right] \\ & \lesssim \underbrace{\inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{H^1(p_{T-t})}^2}_{\text{approximation}} + \underbrace{\bar{B} \mathfrak{R}_n(\mathcal{H}_t) + \frac{\lambda \bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t)}_{\text{generalization}} + \underbrace{\frac{\lambda^2 d^2 \bar{B}^2}{\sigma_{T-t}^8}}_{\text{regularization}}, \\ & \mathbb{E} \left[\|\nabla \hat{h}_t^\lambda - \nabla h_t^*\|_{L^2(p_{T-t})}^2 \right] \\ & \lesssim \underbrace{\frac{1}{\lambda} \inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{H^1(p_{T-t})}^2}_{\text{approximation}} + \underbrace{\frac{\bar{B}}{\lambda} \mathfrak{R}_n(\mathcal{H}_t) + \frac{\bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t)}_{\text{generalization}} + \underbrace{\frac{\lambda d^2 \bar{B}^2}{\sigma_{T-t}^8}}_{\text{regularization}}, \end{aligned}$$

where the notation $\lesssim$ hides absolute constants.

Proof of Lemma 4.2. We first provide a proof sketch of this lemma.

- (i) In this step, we establish the following equality of the gradient penalized empirical risk minimizer $\hat{h}_t^\lambda$ based on the variation inequality:

$$\begin{aligned}
 & \underbrace{\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2\right]}_{\text{(I)}} + \lambda \underbrace{\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \|\nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2\right]}_{\text{(II)}} \\
 (C.5) \quad & \leq \|h_t - h_t^*\|_{L^2(p_{T-t})}^2 + \lambda \|\nabla h_t - \nabla h_t^*\|_{L^2(p_{T-t})}^2 + V_t^2 \\
 & \quad + 2\lambda \underbrace{\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \langle \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle\right]}_{\text{(III)}}.
 \end{aligned}$$

- (ii) In this step, we provide the generalization error bounds for terms (I) and (II) in (C.5):

$$\begin{aligned}
 & \mathbb{E}[\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2] + V_t^2 \\
 (C.6) \quad & \leq \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2\right] + 12\bar{B}\mathfrak{R}_n(\mathcal{H}_t),
 \end{aligned}$$

$$\begin{aligned}
 & \mathbb{E}[\|\nabla \hat{h}_t^\lambda - \nabla h_t^*\|_{L^2(p_{T-t})}^2] \\
 (C.7) \quad & \leq \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \|\nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2\right] + \frac{16\bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t).
 \end{aligned}$$

- (iii) In this step, we estimate term (III) in (C.5):

$$\begin{aligned}
 & \lambda \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \langle \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle\right] - \frac{1}{2} \mathbb{E}[\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2] \\
 (C.8) \quad & \leq \frac{60\lambda^2 d^2 \bar{B}^2}{\sigma_{T-t}^8} + \frac{1}{2} \|h_t - h_t^*\|_{L^2(p_{T-t})}^2 + \frac{8\lambda \bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t),
 \end{aligned}$$

Step 1. Since $\mathcal{H}_t$ is convex, for each $h_t \in \mathcal{H}_t$, we have $\hat{h}_t^\lambda + \varepsilon(h_t - \hat{h}_t^\lambda) \in \mathcal{H}_t$ for each $\varepsilon \in (0, 1)$. Since that $\hat{h}_t^\lambda$ is the minimizer of the empirical risk $\hat{\mathcal{J}}_t^\lambda$ over $\mathcal{H}_t$, it follows that

$$\frac{1}{\varepsilon} \left\{ \hat{\mathcal{J}}_t^\lambda(\hat{h}_t^\lambda + \varepsilon(h_t - \hat{h}_t^\lambda)) - \hat{\mathcal{J}}_t^\lambda(\hat{h}_t^\lambda) \right\} \geq 0.$$

Taking the limit as $\varepsilon \rightarrow 0^+$ on both sides of this inequality yields

$$\begin{aligned}
 & \delta \hat{\mathcal{J}}_t^\lambda(\hat{h}_t^\lambda, h_t - \hat{h}_t^\lambda) \\
 & := \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \left\{ \hat{\mathcal{J}}_t^\lambda(\hat{h}_t^\lambda + \varepsilon(h_t - \hat{h}_t^\lambda)) - \hat{\mathcal{J}}_t^\lambda(\hat{h}_t^\lambda) \right\} \\
 & = \frac{d}{d\varepsilon} \hat{\mathcal{J}}_t^\lambda(\hat{h}_t^\lambda + \varepsilon(h_t - \hat{h}_t^\lambda)) \Big|_{\varepsilon=0} \\
 & = \frac{d}{d\varepsilon} \hat{\mathcal{J}}_t^\lambda(\hat{h}_t^\lambda + \varepsilon(h_t - \hat{h}_t^\lambda)) \Big|_{\varepsilon=0} + \lambda \frac{d}{d\varepsilon} \hat{\mathcal{G}}_t^\lambda(\hat{h}_t^\lambda + \varepsilon(h_t - \hat{h}_t^\lambda)) \Big|_{\varepsilon=0} \\
 & = \frac{2}{n} \sum_{i=1}^n \left\{ (\hat{h}_t^\lambda(\mathbf{X}_t^i) - w(\mathbf{X}_0^i))(h_t(\mathbf{X}_t^i) - \hat{h}_t^\lambda(\mathbf{X}_t^i)) + \lambda \langle \nabla \hat{h}_t^\lambda(\mathbf{X}_t^i), \nabla h_t(\mathbf{X}_t^i) - \nabla \hat{h}_t^\lambda(\mathbf{X}_t^i) \rangle \right\} \\
 & \geq 0,
 \end{aligned}$$

where $\delta \widehat{\mathcal{J}}_t^\lambda$ denotes the first variation (Giaquinta and Hildebrandt, 2004) of $\widehat{\mathcal{J}}_t^\lambda$. Consequently,

$$\begin{aligned}
 (C.9) \quad & \underbrace{\frac{2}{n} \sum_{i=1}^n (\widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))(h_t(\mathbf{X}_{T-t}^i) - \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i))}_{(i)} \\
 & + \underbrace{\frac{2\lambda}{n} \sum_{i=1}^n \langle \nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle}_{(ii)} \\
 & \geq -\frac{2\lambda}{n} \sum_{i=1}^n \nabla \langle h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle.
 \end{aligned}$$

For term (i) in the right-hand side of (C.9), we have

$$\begin{aligned}
 & \frac{1}{n} \sum_{i=1}^n (\widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))(h_t(\mathbf{X}_{T-t}^i) - \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i)) \\
 & = \frac{1}{n} \sum_{i=1}^n (\widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))(h_t(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i) + w(\mathbf{X}_0^i) - \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i)) \\
 & = \frac{1}{n} \sum_{i=1}^n (\widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))(h_t(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i)) - \frac{1}{n} \sum_{i=1}^n (\widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2 \\
 (C.10) \quad & \leq \frac{1}{2n} \sum_{i=1}^n (h_t(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2 - \frac{1}{2n} \sum_{i=1}^n (\widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2,
 \end{aligned}$$

where the inequality holds from $ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2$. By a similar argument, we find

$$\begin{aligned}
 & \frac{1}{n} \sum_{i=1}^n \langle \nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle \\
 & = \frac{1}{n} \sum_{i=1}^n \langle \nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i) \rangle \\
 & \quad - \frac{1}{n} \sum_{i=1}^n \|\nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2 \\
 (C.11) \quad & \leq \frac{1}{2n} \sum_{i=1}^n \|\nabla h_t(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2 - \frac{1}{2n} \sum_{i=1}^n \|\nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2.
 \end{aligned}$$

Substituting (C.10) and (C.11) into (C.9) yields

$$\begin{aligned}
 & \frac{1}{n} \sum_{i=1}^n (\widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2 + \frac{\lambda}{n} \sum_{i=1}^n \|\nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2 \\
 & \leq \frac{1}{n} \sum_{i=1}^n (h_t(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2 + \frac{\lambda}{n} \sum_{i=1}^n \|\nabla h_t(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2 \\
 & \quad + \frac{2\lambda}{n} \sum_{i=1}^n \langle \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \widehat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle.
 \end{aligned}$$

By taking expectation with respect to $\{(\mathbf{X}_0^i, \mathbf{X}_{T-t}^i)\}_{i=1}^n$, it follows that for each $h_t \in \mathcal{H}_t$,

$$\begin{aligned}
 & \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2 \right] + \lambda \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \|\nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i)\|_2^2 \right] \\
 & \leq \mathbb{E} [(h_t(\mathbf{X}_t) - w(\mathbf{X}_0))^2] + \lambda \mathbb{E}_{\mathbf{X}_t} [\|\nabla h_t(\mathbf{X}_t) - \nabla h_t^*(\mathbf{X}_t)\|_2^2] \\
 & \quad + 2\lambda \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \langle \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle \right] \\
 & \leq \|h_t - h_t^*\|_{L^2(p_{T-t})}^2 + \lambda \|\nabla h_t - \nabla h_t^*\|_{L^2(p_{T-t})}^2 + V_t^2 \\
 & \quad + 2\lambda \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \langle \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle \right],
 \end{aligned}$$

where the second inequality follows from Proposition 3.4. This completes the proof of (C.5).

Step 2. It follows from $h_t^*(\mathbf{X}_{T-t}^i) = \mathbb{E}[w(\mathbf{X}_0^i)|\mathbf{X}_{T-t}^i]$ that

$$\begin{aligned}
 & \frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2 \\
 & = \frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i) + \mathbb{E}[w(\mathbf{X}_0^i)|\mathbf{X}_{T-t}^i] - w(\mathbf{X}_0^i))^2 \\
 & = \frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 + \frac{1}{n} \sum_{i=1}^n (\mathbb{E}[w(\mathbf{X}_0^i)|\mathbf{X}_{T-t}^i] - w(\mathbf{X}_0^i))^2 \\
 & \quad + \frac{2}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))(\mathbb{E}[w(\mathbf{X}_0^i)|\mathbf{X}_{T-t}^i] - w(\mathbf{X}_0^i)).
 \end{aligned}$$

Taking expectation with respect to $\{(\mathbf{X}_0^i, \mathbf{X}_{T-t}^i)\}_{i=1}^n$ implies

$$\begin{aligned}
 & \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - w(\mathbf{X}_0^i))^2 \right] \\
 & = \underbrace{\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right]}_{(i)} + V_t^2 \\
 & \quad + 2 \underbrace{\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) (\mathbb{E}[w(\mathbf{X}_0^i)|\mathbf{X}_{T-t}^i] - w(\mathbf{X}_0^i)) \right]}_{(ii)},
 \end{aligned} \tag{C.12}$$

where V_t^2 is defined in Proposition 3.4, and we used the fact that h_t^* is independent of $\{(\mathbf{X}_0^i, \mathbf{X}_{T-t}^i)\}_{i=1}^n$. For term (i) in (C.12), using Lemma C.7 yields

$$\mathbb{E} [\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2] \leq \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right] + 8\bar{B}\mathfrak{R}_n(\mathcal{H}_t). \tag{C.13}$$

For term (ii) in (C.12), by applying Lemma C.11, we find

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) (w(\mathbf{X}_0^i) - \mathbb{E}[w(\mathbf{X}_0^i)|\mathbf{X}_{T-t}^i]) \right] \leq 2\bar{B}\mathfrak{R}(\mathcal{H}_t). \tag{C.14}$$

Plugging (C.13) and (C.14) into (C.12) obtains (C.6). The inequality (C.7) is a direct conclusion of Lemma C.8.

Step 4. It follows from Lemma C.9 that for each $\xi > 0$,

$$\begin{aligned}
& \lambda \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \langle \nabla h_t^*(\mathbf{X}_{T-t}^i), \nabla h_t(\mathbf{X}_{T-t}^i) - \nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \rangle \right] \\
& \leq \lambda \mathbb{E} [(\nabla h_t^*, \nabla h_t - \nabla \hat{h}_t^\lambda)_{L^2(p_{T-t})}] + \frac{8\lambda \bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t) \\
& \leq \frac{30}{\xi} \frac{\lambda d^2 \bar{B}^2}{\sigma_{T-t}^8} + \xi \lambda \mathbb{E} [\|h_t - \hat{h}_t^\lambda\|_{L^2(p_{T-t})}^2] + \frac{8\lambda \bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t) \\
& \leq \frac{30}{\xi} \frac{\lambda d^2 \bar{B}^2}{\sigma_{T-t}^8} + \xi \lambda \|h_t - h_t^*\|_{L^2(p_{T-t})}^2 + \xi \lambda \mathbb{E} [\|h_t^* - \hat{h}_t^\lambda\|_{L^2(p_{T-t})}^2] + \frac{8\lambda \bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t),
\end{aligned}$$

where the first inequality is due to Lemma C.9, the second inequality follows Lemma C.6, and the third inequality comes from the triangular inequality. Finally, setting $\xi = \frac{1}{2\lambda}$ yields (C.8).

Finally, combining (C.5), (C.6), (C.7), and (C.8) completes the proof. $\square$

C.3 Auxiliary lemmas for the oracle inequality.

Lemma C.6. *Suppose Assumptions 1 and 2 hold. Assume that $|v_t(\mathbf{x})| < \infty$ for each $\mathbf{x} \in \mathbb{R}^d$. Then for each $\xi > 0$,*

$$|(\nabla h_t^*, \nabla v_t)_{L^2(p_{T-t})}| \leq \frac{30}{\xi} \frac{d^2 \bar{B}^2}{\sigma_{T-t}^8} + \xi \|v_t\|_{L^2(p_{T-t})}^2.$$

Proof of Lemma C.6. By a direct calculation, we have

$$\begin{aligned}
& -(\nabla h_t^*, \nabla v_t)_{L^2(p_{T-t})} \\
& = - \int_{\mathbb{R}^d} \langle \nabla h_t^*(\mathbf{x}), \nabla v_t(\mathbf{x}) \rangle p_{T-t}(\mathbf{x}) \, d\mathbf{x} \\
(C.15) \quad & = - \underbrace{\int_{\mathbb{R}^d} \nabla \cdot (\nabla h_t^*(\mathbf{x}) p_{T-t}(\mathbf{x}) v_t(\mathbf{x})) \, d\mathbf{x}}_{(i)} + \underbrace{\int_{\mathbb{R}^d} \nabla \cdot (\nabla h_t^*(\mathbf{x}) p_t(\mathbf{x})) v_t(\mathbf{x}) \, d\mathbf{x}}_{(ii)}.
\end{aligned}$$

Before addressing the term (i) in (C.15), we first restrict this integral on a d -dimensional ball $B(\mathbf{0}, R) := \{\mathbf{x} : \|\mathbf{x}\|_2 \leq R\}$:

$$(C.16) \quad \int_{B(\mathbf{0}, R)} \nabla \cdot (\nabla h_t^*(\mathbf{x}) p_{T-t}(\mathbf{x}) v_t(\mathbf{x})) \, d\mathbf{x} = \int_{\partial B(\mathbf{0}, R)} \langle \nabla h_t^*(\mathbf{x}), \mathbf{n} \rangle p_{T-t}(\mathbf{x}) v_t(\mathbf{x}) \, dS,$$

where we used Gauss's divergence theorem. Here $\partial B(\mathbf{0}, R)$ is the boundary of $B(\mathbf{0}, R)$, and

$\mathbf{n}$ is the outward-pointing unit normal vector on $\partial B(\mathbf{0}, R)$. Furthermore,

$$\begin{aligned}
 & \left| \int_{\partial B(\mathbf{0}, R)} \langle \nabla h_t^*(\mathbf{x}), \mathbf{n} \rangle p_{T-t}(\mathbf{x}) v_t(\mathbf{x}) dS \right| \\
 & \leq \int_{\partial B(\mathbf{0}, R)} |\langle \nabla h_t^*(\mathbf{x}), \mathbf{n} \rangle p_{T-t}(\mathbf{x}) v_t(\mathbf{x})| dS \\
 & \leq \int_{\partial B(\mathbf{0}, R)} \sup_{\mathbf{x} \in B(\mathbf{0}, R)} \|\nabla h_t^*(\mathbf{x})\|_2 \sup_{\mathbf{x} \in B(\mathbf{0}, R)} |v_t(\mathbf{x})| p_{T-t}(\mathbf{x}) dS \\
 & \leq \frac{2\sqrt{d}\bar{B}}{(2\pi)^{d/2} \sigma_t^{d+2}} \sup_{\mathbf{x} \in B(\mathbf{0}, R)} |v_t(\mathbf{x})| \exp\left(-\frac{(R-1)^2}{2\sigma_{T-t}^2}\right) \int_{\partial B(\mathbf{0}, R)} dS \\
 & = \frac{2\sqrt{d}A_{d-1}\bar{B}}{(2\pi)^{d/2} \sigma_{T-t}^{d+2}} \sup_{\mathbf{x} \in B(\mathbf{0}, R)} |v_t(\mathbf{x})| \exp\left(-\frac{(R-1)^2}{2\sigma_{T-t}^2}\right) R^{d-1},
 \end{aligned}$$

where A_{d-1} is the surface area of the unit ball in d -dimensional space, the first inequality is due to Jensen's inequality, the second inequality holds from $|\langle \nabla h_t^*(\mathbf{x}), \mathbf{n} \rangle| \leq \|\nabla h_t^*(\mathbf{x})\|_2 \|\mathbf{n}\|_2 = \|\nabla h_t^*(\mathbf{x})\|_2$, and the third inequality involves Lemma C.4, and the fact that

$$\begin{aligned}
 p_{T-t}(\mathbf{x}) & \leq \sup_{\mathbf{x}_0 \in B(\mathbf{0}, 1)} \varphi_d(\mathbf{x}; \mu_{T-t}\mathbf{x}_0, \sigma_{T-t}^2 \mathbf{I}_d) \int p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 & \leq \frac{1}{(2\pi)^{d/2} \sigma_{T-t}^d} \exp\left(-\frac{(\|\mathbf{x}\|_2 - 1)^2}{2\sigma_{T-t}^2}\right).
 \end{aligned}$$

Here we used Assumption 1. As a result,

$$\lim_{R \rightarrow \infty} \left| \int_{\partial B(\mathbf{0}, R)} \langle \nabla h_t^*(\mathbf{x}), \mathbf{n} \rangle p_{T-t}(\mathbf{x}) v_t(\mathbf{x}) dS \right| = 0.$$

Combining this equality with (C.16) yields

$$(C.17) \quad \int_{\mathbb{R}^d} \nabla \cdot (\nabla h_t^*(\mathbf{x}) p_{T-t}(\mathbf{x}) v_t(\mathbf{x})) d\mathbf{x} = \lim_{R \rightarrow \infty} \int_{B(\mathbf{0}, R)} \nabla \cdot (\nabla h_t^*(\mathbf{x}) p_{T-t}(\mathbf{x}) v_t(\mathbf{x})) d\mathbf{x} = 0.$$

For the term (ii) in (C.15),

$$\begin{aligned}
 & \int_{\mathbb{R}^d} \nabla \cdot (\nabla h_t^*(\mathbf{x}) p_{T-t}(\mathbf{x}) v_t(\mathbf{x})) d\mathbf{x} \\
 & = \int_{\mathbb{R}^d} (\Delta h_t^*(\mathbf{x}) p_{T-t}(\mathbf{x}) + \nabla h_t^*(\mathbf{x}) \cdot \nabla p_{T-t}(\mathbf{x})) v_t(\mathbf{x}) d\mathbf{x} \\
 (C.18) \quad & = (\Delta h_t^* + \nabla h_t^* \cdot \nabla \log p_{T-t}, v_t)_{L^2(p_{T-t})}.
 \end{aligned}$$

Substituting (C.17) and (C.18) into (C.15) yields

$$(C.19) \quad -(\nabla h_t^*, \nabla v_t)_{L^2(p_{T-t})} = (\Delta h_t^* + \nabla h_t^* \cdot \nabla \log p_{T-t}, v_t)_{L^2(p_{T-t})}.$$

Then using Cauchy-Schwarz inequality and triangular inequality implies

$$\begin{aligned}
 & |(\nabla h_t^*, \nabla v_t)_{L^2(p_{T-t})}| \\
 & \leq \{ \|\Delta h_t^*\|_{L^2(p_{T-t})} + \|\nabla h_t^* \cdot \nabla \log p_{T-t}\|_{L^2(p_{T-t})} \} \|v_t\|_{L^2(p_{T-t})} \\
 & \leq \frac{1}{2\xi} \|\Delta h_t^*\|_{L^2(p_{T-t})}^2 + \frac{1}{2\xi} \|\nabla h_t^* \cdot \nabla \log p_{T-t}\|_{L^2(p_{T-t})}^2 + \xi \|v_t\|_{L^2(p_{T-t})}^2 \\
 & \leq \frac{1}{2\xi} \|\Delta h_t^*\|_{L^2(p_{T-t})}^2 + \frac{1}{2\xi} \|\nabla h_t^*\|_{L^2(p_{T-t})}^2 \|\nabla \log p_{T-t}\|_{L^2(p_{T-t})}^2 + \xi \|v_t\|_{L^2(p_{T-t})}^2 \\
 (C.20) \quad & \leq \frac{18 d^2 \bar{B}^2}{\xi \sigma_t^8} + \frac{2 d \bar{B}^2}{\xi \sigma_t^4} \|\nabla \log p_{T-t}\|_{L^2(p_{T-t})}^2 + \xi \|v_t\|_{L^2(p_{T-t})}^2,
 \end{aligned}$$

where the first inequality holds from (C.19) and Cauchy-Schwarz inequality, the second inequality is due to the inequality $ab \leq \frac{1}{2\epsilon}a + \frac{\epsilon}{2}b$, and the last inequality invokes Lemma C.4 and C.5. It remains to estimate the $L^2(p_{T-t})$ -norm of the score $\nabla \log p_{T-t}$. Indeed,

$$\begin{aligned}
 & \|\nabla \log p_{T-t}\|_{L^2(p_{T-t})}^2 \\
 &= \int \left\| \frac{\mathbf{x}}{\sigma_{T-t}^2} - \frac{\mu_{T-t}}{\sigma_{T-t}^2} \mathbb{E}[\mathbf{X}_0 | \mathbf{X}_{T-t} = \mathbf{x}] \right\|_2^2 p_{T-t}(\mathbf{x}) d\mathbf{x} \\
 &\leq \frac{2}{\sigma_{T-t}^4} \mathbb{E}[\|\mathbf{X}_{T-t}\|_2^2] + \frac{2d\mu_{T-t}^2}{\sigma_{T-t}^4} \\
 &= \frac{2}{\sigma_{T-t}^4} \left\{ \mu_{T-t}^2 \mathbb{E}[\|\mathbf{X}_0\|_2^2] + 2\mu_{T-t}\sigma_{T-t} \mathbb{E}[\langle \mathbf{X}_0, \boldsymbol{\varepsilon} \rangle] + \sigma_{T-t}^2 \mathbb{E}[\|\boldsymbol{\varepsilon}\|_2^2] \right\} + \frac{2d\mu_{T-t}^2}{\sigma_{T-t}^4} \\
 \text{(C.21)} \quad &= \frac{2}{\sigma_{T-t}^4} \left\{ \mu_{T-t}^2 \mathbb{E}[\|\mathbf{X}_0\|_2^2] + \sigma_{T-t}^2 \mathbb{E}[\|\boldsymbol{\varepsilon}\|_2^2] \right\} + \frac{2d\mu_{T-t}^2}{\sigma_{T-t}^4} \leq \frac{6d}{\sigma_{T-t}^4},
 \end{aligned}$$

where the first equality is owing to Lemma C.2, the first inequality used Assumption 1, and the second and third equalities hold from $\mathbf{X}_{T-t} \stackrel{d}{=} \mu_{T-t}\mathbf{X}_0 + \sigma_{T-t}\boldsymbol{\varepsilon}$ where $\mathbf{X}_0$ is independent $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. The last inequality also use Assumption 1. Substituting (C.21) into (C.20) completes the proof. $\square$

According to the standard techniques of symmetrization (Mohri et al., 2018, Theorem 3.3), we have the following generalization bounds.

Lemma C.7. *Suppose Assumptions 1 and 2 hold. Then*

$$\mathbb{E} \left[\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2 \right] \leq \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right] + 8\bar{B}\mathfrak{R}_n(\mathcal{H}),$$

where the expectation is taken with respect to $\mathbf{X}_{T-t}^1, \dots, \mathbf{X}_{T-t}^n \sim^{\text{i.i.d.}} p_{T-t}$.

Proof of Lemma C.7. Let $\mathbf{X}_{T-t}^{1'}, \dots, \mathbf{X}_{T-t}^{n'}$ be independent copies of $\mathbf{X}_{T-t}^1, \dots, \mathbf{X}_{T-t}^n$. Let $\varepsilon_1, \dots, \varepsilon_n$ be a set of i.i.d. Rademacher variables, which are independent of $\mathbf{X}_{T-t}^{1'}, \dots, \mathbf{X}_{T-t}^{n'}$

and $\mathbf{X}_{T-t}^1, \dots, \mathbf{X}_{T-t}^n$. Since $\hat{h}_t^\lambda \in \mathcal{H}_t$, it follows that

$$\begin{aligned}
 & \mathbb{E} \left[\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2 - \frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right] \\
 &= \mathbb{E} \left[\mathbb{E}[(\hat{h}_t^\lambda(\mathbf{X}_{T-t}) - h_t^*(\mathbf{X}_{T-t}))^2 | \mathbf{X}_{T-t}^{1:n}] - \frac{1}{n} \sum_{i=1}^n (\hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right] \\
 &\leq \mathbb{E} \left[\sup_{h_t \in \mathcal{H}_t} \mathbb{E}[(h_t(\mathbf{X}_{T-t}) - h_t^*(\mathbf{X}_{T-t}))^2] - \frac{1}{n} \sum_{i=1}^n (h_t(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right] \\
 &= \mathbb{E} \left[\sup_{h_t \in \mathcal{H}_t} \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (h_t(\mathbf{X}_{T-t}^{i'}) - h_t^*(\mathbf{X}_{T-t}^{i'}))^2 \right] - \frac{1}{n} \sum_{i=1}^n (h_t(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right] \\
 &\leq \mathbb{E} \left[\sup_{h_t \in \mathcal{H}_t} \frac{1}{n} \sum_{i=1}^n \left\{ (h_t(\mathbf{X}_{T-t}^{i'}) - h_t^*(\mathbf{X}_{T-t}^{i'}))^2 - (h_t(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right\} \right] \\
 &= \mathbb{E} \left[\sup_{h_t \in \mathcal{H}_t} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \left\{ (h_t(\mathbf{X}_{T-t}^{i'}) - h_t^*(\mathbf{X}_{T-t}^{i'}))^2 - (h_t(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right\} \right] \\
 &= 2\mathbb{E} \left[\sup_{h_t \in \mathcal{H}_t} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (h_t(\mathbf{X}_{T-t}^i) - h_t^*(\mathbf{X}_{T-t}^i))^2 \right] \leq 8\bar{B}\mathfrak{R}_n(\mathcal{H}_t),
 \end{aligned}$$

where the second inequality holds from Jensen's inequality, and last inequality is due to Ledoux-Talagrand contraction inequality (Mohri et al., 2018, Lemma 5.7) and Lemma C.1. This completes the proof. $\square$

By a similar argument as Lemma C.7, we have the following generalization bounds.

Lemma C.8. *Suppose Assumptions 1 and 2 hold. Then*

$$\mathbb{E} \left[\|\nabla \hat{h}_t^\lambda - \nabla h_t^*\|_{L^2(p_{T-t})}^2 \right] \leq \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) - \nabla h_t^*(\mathbf{X}_{T-t}^i))^2 \right] + \frac{16\bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t),$$

where the expectation is taken with respect to $\mathbf{X}_{T-t}^1, \dots, \mathbf{X}_{T-t}^i \sim \text{i.i.d. } p_{T-t}$, and $\mathfrak{R}_n(\nabla \mathcal{H}_t) := \sum_{k=1}^d \mathfrak{R}_n(D_k \mathcal{H}_t)$.

Lemma C.9. *Suppose Assumptions 1 and 2 hold. Then*

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \nabla h_t^*(\mathbf{X}_{T-t}^i) \cdot \nabla \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \right] \leq \mathbb{E} \left[(\nabla h_t^*, \nabla \hat{h}_t^\lambda)_{L^2(p_{T-t})} \right] + \frac{8\bar{B}}{\sigma_{T-t}^2} \mathfrak{R}_n(\nabla \mathcal{H}_t),$$

where the expectation is taken with respect to $\mathbf{X}_{T-t}^1, \dots, \mathbf{X}_{T-t}^i \sim \text{i.i.d. } p_{T-t}$, and $\mathfrak{R}_n(\nabla \mathcal{H}_t) := \sum_{k=1}^d \mathfrak{R}_n(D_k \mathcal{H}_t)$.

The following lemma is an extension of Bartlett and Mendelson (2002, Lemma 4).

Lemma C.10. *Let $\mathbf{z} = (z_1, \dots, z_n) \in \mathcal{Z} \subseteq \mathbb{R}^n$. Let $\xi_1, \dots, \xi_n$ be a sequence of i.i.d. random variables with $|\xi_i| < K$ and $\mathbb{E}[\xi_i] = 0$ for each $1 \leq i \leq n$. Then it follows that*

$$\mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \frac{1}{n} \sum_{i=1}^n \xi_i z_i \right] \leq 2K \mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i z_i \right],$$

where $\varepsilon_1, \dots, \varepsilon_n$ is a sequence of i.i.d. Rademacher variables.

Proof of Lemma C.10. The proof relies on the symmetrization technique. Let $\xi'_1, \dots, \xi'_n$ be independent copies of $\xi_1, \dots, \xi_n$. It follows that

$$\begin{aligned} \mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \frac{1}{n} \sum_{i=1}^n \xi_i z_i \right] &= \mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\xi_i - \xi'_i) z_i \middle| \xi_1, \dots, \xi_n \right] \right] \\ &\leq \mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \frac{1}{n} \sum_{i=1}^n (\xi_i - \xi'_i) z_i \right] = \mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (\xi_i - \xi'_i) z_i \right] \\ &\leq 2 \mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \xi_i z_i \right] \leq 2K \mathbb{E} \left[\sup_{\mathbf{z} \in \mathcal{Z}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i z_i \right], \end{aligned}$$

where the first equality due to $\mathbb{E}[\xi'_i] = 0$, the first inequality holds from Jensen's inequality, and the second equality follows from the fact that distribution of $(\xi_i - \xi'_i)$ is symmetric around zero, so it has the same distribution as $\varepsilon_i(\xi_i - \xi'_i)$. The second inequality comes from the triangular inequality for the supremum, and we used the fact that ξ_i and ξ'_i are identically distributed. The last inequality invokes Ledoux-Talagrand contraction inequality (Mohri et al., 2018, Lemma 5.7) and $\max_{1 \leq i \leq n} |\xi_i| \leq K$. This completes the proof. $\square$

Lemma C.11. Suppose Assumptions 1 and 2 hold. Then

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) (w(\mathbf{X}_0^i) - \mathbb{E}[w(\mathbf{X}_0^i) | \mathbf{X}_{T-t}^i]) \middle| \mathbf{X}_{T-t}^{1:n} \right] \leq 2\bar{B} \mathfrak{R}(\mathcal{H}_t | \mathbf{X}_{T-t}^{1:n}),$$

where the expectation is taken with respect to $\mathbf{X}_{T-t}^1, \dots, \mathbf{X}_{T-t}^i \sim^{\text{i.i.d.}} p_{T-t}$.

Proof of Lemma C.11. Define a sequence of random variables

$$\xi_i = w(\mathbf{X}_0^i) - \mathbb{E}[w(\mathbf{X}_0^i) | \mathbf{X}_{T-t}^i].$$

It is apparent that $\mathbb{E}[\xi_i | \mathbf{X}_{T-t}^i] = 0$, and $|\xi_i| \leq \bar{B}$. Using Lemma C.10 yields

$$\begin{aligned} \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \xi_i \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \middle| \mathbf{X}_{T-t}^{1:n} \right] &\leq 2\bar{B} \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \varepsilon_i \hat{h}_t^\lambda(\mathbf{X}_{T-t}^i) \middle| \mathbf{X}_{T-t}^{1:n} \right] \\ &\leq 2\bar{B} \mathbb{E} \left[\sup_{h_t \in \mathcal{H}_t} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h_t(\mathbf{X}_{T-t}^i) \middle| \mathbf{X}_{T-t}^{1:n} \right] \\ &= 2\bar{B} \mathfrak{R}(\mathcal{H}_t | \mathbf{X}_{T-t}^{1:n}), \end{aligned}$$

where $\varepsilon_1, \dots, \varepsilon_n$ is a sequence of i.i.d. Rademacher variables. Here the first inequality follows from Lemma C.10, and the second inequality is due to the fact that $\hat{h}_t^\lambda \in \mathcal{H}_t$, the second inequality holds from Lemma C.10. This completes the proof. $\square$

C.4 Convergence rate of the Doob's guidance estimation.

Lemma C.12. Suppose Assumptions 1 and 2 hold. Let the hypothesis class $\mathcal{H}_t$ be defined as (4.3) with $L \leq C \log N$ and $S \leq N^d$, then there exists $h_t \in \mathcal{H}_t$ such that

$$\begin{aligned} \|h_t - h_t^*\|_{L^\infty(B(\mathbf{0}, R))} &\leq \frac{CR^2}{\sigma_{T-t}^4 N^2}, \\ \|\nabla h_t - \nabla h_t^*\|_{L^\infty(B(\mathbf{0}, R))} &\leq \frac{CR}{\sigma_{T-t}^4 N}, \end{aligned}$$

where C is a constant independent of N and R .

Proof of Lemma C.12. We first rescale the target function h_t^* to $B(\mathbf{0}, 1)$ by $g_t^*(\mathbf{z}) := h_t^*(R\mathbf{z})$. According to Ding et al. (2025, Lemma 6), there exists $g_t \in N(L, S)$ such that

$$\begin{aligned}\|g_t - g_t^*\|_{L^\infty(B(\mathbf{0}, 1))} &\leq \frac{C}{N^2} \|g_t^*\|_{C^2(\mathbb{R}^d)}, \\ \|\nabla g_t - \nabla g_t^*\|_{L^\infty(B(\mathbf{0}, 1))} &\leq \frac{C}{N} \|g_t^*\|_{C^2(\mathbb{R}^d)},\end{aligned}$$

where C is a constant only depend on d . Note that $D_k g_t^*(\mathbf{z}) = R D_k h_t^*(R\mathbf{z})$ for each $1 \leq k \leq d$, and $D_{k\ell}^2 g_t^*(\mathbf{z}) = R^2 D_{k\ell}^2 h_t^*(R\mathbf{z})$ for each $1 \leq k, \ell \leq d$. Thus

$$\begin{aligned}\|g_t(R^{-1}\cdot) - h_t^*\|_{L^\infty(B(\mathbf{0}, R))} &= \|g_t(R^{-1}\cdot) - g_t^*(R^{-1}\cdot)\|_{L^\infty(B(\mathbf{0}, 1))} \leq \frac{CR^2}{N^2} \|h_t^*\|_{C^2(\mathbb{R}^d)}, \\ \|\nabla g_t(R^{-1}\cdot) - \nabla h_t^*\|_{L^\infty(B(\mathbf{0}, R))} &= \frac{1}{R} \|\nabla g_t - \nabla g_t^*\|_{L^\infty(B(\mathbf{0}, 1))} \leq \frac{CR}{N} \|h_t^*\|_{C^2(\mathbb{R}^d)}.\end{aligned}$$

Set $h_t := g_t(R^{-1}\cdot)$, and using Lemmas C.1, C.4, and C.5 complete the proof. $\square$

Lemma C.13 (Approximation error). *Suppose Assumptions 1 and 2 hold. Let the hypothesis class $\mathcal{H}_t$ be defined as (4.3) with $L \leq C \log N$ and $S \leq N^d$, then*

$$\inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{H^1(p_{T-t})}^2 \leq \frac{CR^4}{\sigma_{T-t}^8 N^2} + \frac{C}{\sigma_{T-t}^4} \exp\left(-\frac{R^2}{4d\mu_t^2 + 8\sigma_t^2}\right),$$

where C is a constant independent of t , N and R . Further, by setting $R^2 = (4d\mu_t^2 + 8\sigma_t^2) \log N^2$, we have

$$\inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{H^1(p_{T-t})}^2 \leq \frac{C' \log^2 N}{\sigma_{T-t}^8 N^2},$$

where C' is a constant independent of t and N .

Proof of Lemma C.13. It is straightforward that for each $R \geq 1$,

$$\begin{aligned}(C.22) \quad &\|h_t - h_t^*\|_{H^1(p_{T-t})}^2 \\ &= \underbrace{\int \left((h_t(\mathbf{x}) - h_t^*(\mathbf{x}))^2 + \|\nabla h_t(\mathbf{x}) - \nabla h_t^*(\mathbf{x})\|_2^2 \right) \mathbb{1}_{\{\|\mathbf{x}\|_2 \leq R\}} p_{T-t}(\mathbf{x}) \, d\mathbf{x}}_{(i)} \\ &\quad + \underbrace{\int \left((h_t(\mathbf{x}) - h_t^*(\mathbf{x}))^2 + \|\nabla h_t(\mathbf{x}) - \nabla h_t^*(\mathbf{x})\|_2^2 \right) \mathbb{1}_{\{\|\mathbf{x}\|_2 > R\}} p_{T-t}(\mathbf{x}) \, d\mathbf{x}}_{(ii)}.\end{aligned}$$

For term (i) in (C.22), we have

$$\begin{aligned}(C.23) \quad &\int \left((h_t(\mathbf{x}) - h_t^*(\mathbf{x}))^2 + \|\nabla h_t(\mathbf{x}) - \nabla h_t^*(\mathbf{x})\|_2^2 \right) \mathbb{1}_{\{\|\mathbf{x}\|_2 \leq R\}} p_{T-t}(\mathbf{x}) \, d\mathbf{x} \\ &\leq \sup_{\|\mathbf{x}\|_2 \leq R} \left((h_t(\mathbf{x}) - h_t^*(\mathbf{x}))^2 + \|\nabla h_t(\mathbf{x}) - \nabla h_t^*(\mathbf{x})\|_2^2 \right) \leq \frac{CR^4}{\sigma_{T-t}^8 N^2},\end{aligned}$$

where the second inequality holds from Lemma C.12. For term (ii) in (C.22), we have

$$\begin{aligned}(C.24) \quad &\int \|h_t(\mathbf{x}) - h_t^*(\mathbf{x})\|_2^2 \mathbb{1}_{\{\|\mathbf{x}\|_2 > R\}} p_{T-t}(\mathbf{x}) \, d\mathbf{x} \\ &\leq 4 \left(\bar{B}^2 + \frac{4\bar{B}^2}{\sigma_{T-t}^4} \right) \Pr\{\|\mathbf{X}_{T-t}\|_2 > R\} \\ &\leq 2^{d+3} \left(\bar{B}^2 + \frac{4\bar{B}^2}{\sigma_{T-t}^4} \right) \exp\left(-\frac{R^2}{4d\mu_t^2 + 8\sigma_t^2}\right),\end{aligned}$$

where the first inequality holds from Lemmas C.1 and C.4, and the second inequality is due to Lemma E.1. Combining (C.22), (C.23), and (C.24) yields

$$\inf_{h_t \in \mathcal{H}_t} \|h_t - h_t^*\|_{H^1(p_{T-t})}^2 \leq \frac{CR^4}{\sigma_{T-t}^8 N^2} + \frac{C}{\sigma_{T-t}^4} \exp\left(-\frac{R^2}{4d\mu_t^2 + 8\sigma_t^2}\right).$$

Setting $R^2 = (4d\mu_t^2 + 8\sigma_t^2) \log N^2$ completes the proof. $\square$

Lemma C.14 (Generalization error). *Suppose Assumptions 1 and 2 hold. Let the hypothesis class $\mathcal{H}_t$ be defined as (4.3), then*

$$\begin{aligned} \mathfrak{R}_n(\mathcal{H}_t) &\leq c\bar{B} \left(\frac{LS \log(S)}{n \log^{-1} n} \right)^{\frac{1}{2}}, \\ \mathfrak{R}_n(\nabla \mathcal{H}_t) &\leq c \frac{\bar{B}}{\sigma_{T-t}^2} \left(\frac{L^2 S \log(LS)}{n \log^{-1} n} \right)^{\frac{1}{2}}, \end{aligned}$$

where c is an absolute constant.

Proof of Lemma C.14. Define a function class $\widetilde{\mathcal{H}}_t$ as

$$\widetilde{\mathcal{H}}_t := \left\{ h_t \in N(L, S) : \begin{array}{l} \sup_{\mathbf{x} \in \mathbb{R}^d} h_t(\mathbf{x}) \leq \bar{B}, \quad \inf_{\mathbf{x} \in \mathbb{R}^d} h_t(\mathbf{x}) \geq \underline{B}, \\ \max_{1 \leq k \leq d} \sup_{\mathbf{x} \in \mathbb{R}^d} |D_k h_t(\mathbf{x})| \leq 2\sigma_{T-t}^{-2} \bar{B}. \end{array} \right\}.$$

It is apparent that $\mathcal{H}_t = \text{conv}(\widetilde{\mathcal{H}}_t)$, and $\nabla \mathcal{H}_t = \text{conv}(\nabla \widetilde{\mathcal{H}}_t)$. Then applying Bartlett and Mendelson (2002, Theorem 12) implies

$$(C.25) \quad \mathfrak{R}(\mathcal{H}_t | \mathbf{X}_{T-t}^{1:n}) = \mathfrak{R}(\widetilde{\mathcal{H}}_t | \mathbf{X}_{T-t}^{1:n}),$$

$$(C.26) \quad \mathfrak{R}(\nabla \mathcal{H}_t | \mathbf{X}_{T-t}^{1:n}) = \mathfrak{R}(\nabla \widetilde{\mathcal{H}}_t | \mathbf{X}_{T-t}^{1:n}),$$

where $\mathbf{X}_{T-t}^1, \dots, \mathbf{X}_{T-t}^n \sim \text{i.i.d. } p_{T-t}$. Let $\delta > 0$ and $\widetilde{\mathcal{H}}_t^\delta$ be an $L^\infty(\mathbf{X}_{T-t}^{1:n})$ δ -cover of $\widetilde{\mathcal{H}}_t$ satisfying $|\widetilde{\mathcal{H}}_t^\delta| = N(\delta, \widetilde{\mathcal{H}}_t, L^\infty(\mathbf{X}_{T-t}^{1:n}))$. Then for each $h_t \in \widetilde{\mathcal{H}}_t$, there exists $h_t^\delta \in \widetilde{\mathcal{H}}_t^\delta$ such that

$$\frac{1}{n} \sum_{i=1}^n \varepsilon_i h_t(\mathbf{X}_t^i) - \frac{1}{n} \sum_{i=1}^n \varepsilon_i h_t^\delta(\mathbf{X}_t^i) \leq \delta.$$

As a consequence,

$$\begin{aligned} \mathfrak{R}(\widetilde{\mathcal{H}}_t | \mathbf{X}_{T-t}^{1:n}) &= \mathbb{E} \left[\sup_{h_t \in \widetilde{\mathcal{H}}_t} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h_t(\mathbf{X}_t^i) \middle| \mathbf{X}_{T-t}^{1:n} \right] \\ &\leq \mathbb{E} \left[\sup_{h_t^\delta \in \widetilde{\mathcal{H}}_t^\delta} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h_t^\delta(\mathbf{X}_t^i) \middle| \mathbf{X}_{T-t}^{1:n} \right] + \delta \\ &\leq \bar{B} \left(\frac{2 \log |\widetilde{\mathcal{H}}_t^\delta|}{n} \right)^{\frac{1}{2}} + \delta \\ &= \bar{B} \left(\frac{2 \log N(\delta, \widetilde{\mathcal{H}}_t, L^\infty(\mathbf{X}_{T-t}^{1:n}))}{n} \right)^{\frac{1}{2}} + \delta, \end{aligned}$$

where $\varepsilon_1, \dots, \varepsilon_n$ is a sequence of i.i.d. Rademacher variables, the second inequality follows from Massart's lemma (Mohri et al., 2018, Theorem 3.7). Setting $\delta = \bar{B}/\sqrt{n}$ yields

$$(C.27) \quad \mathfrak{R}(\widetilde{\mathcal{H}}_t | \mathbf{X}_{T-t}^{1:n}) \leq \bar{B} \left(\frac{2 \log N(B/\sqrt{n}, \widetilde{\mathcal{H}}_t, L^\infty(\mathbf{X}_{T-t}^{1:n}))}{n} \right)^{\frac{1}{2}} \leq 8\bar{B} \left(\frac{\text{VCdim}(\widetilde{\mathcal{H}}_t)}{n \log^{-1} n} \right)^{\frac{1}{2}},$$

where the last inequality holds from Anthony et al. (1999, Theorem 12.2). Since $\widetilde{\mathcal{H}}_t \subseteq N(L, S)$, using Bartlett et al. (2019, Theorem 7) implies

$$(C.28) \quad \text{VCdim}(\widetilde{\mathcal{H}}_t) \leq \text{VCdim}(N(L, S)) \leq c_0 L S \log(S),$$

where c_0 is an absolute constant. Combining (C.25), (C.27), and (C.28) deduces

$$\mathfrak{R}_n(\mathcal{H}_t) = \mathbb{E}[\mathfrak{R}(\mathcal{H}_t | \mathbf{X}_{T-t}^{1:n})] \leq c_1 \bar{B} \left(\frac{L S \log(S)}{n \log^{-1} n} \right)^{\frac{1}{2}},$$

where c_1 is an absolute constant. By a similar argument as (C.27), we have

$$(C.29) \quad \mathfrak{R}(\nabla \widetilde{\mathcal{H}}_t | \mathbf{X}_{T-t}^{1:n}) \leq 16 \frac{\bar{B}}{\sigma_{T-t}^2} \left(\frac{\text{VCdim}(\nabla \widetilde{\mathcal{H}}_t)}{n \log^{-1} n} \right)^{\frac{1}{2}}.$$

Using, we have $\nabla \widetilde{\mathcal{H}}_t \subseteq N(c_2 L, c_3 L S)$, where c_2 and c_3 are absolute constants. Using Bartlett et al. (2019, Theorem 7) implies

$$(C.30) \quad \text{VCdim}(\nabla \widetilde{\mathcal{H}}_t) \leq c_4 L^2 S \log(L S),$$

where c_4 is an absolute constant. Combining (C.26), (C.29), and (C.30) deduces

$$\mathfrak{R}_n(\nabla \mathcal{H}_t) = \mathbb{E}[\mathfrak{R}(\nabla \mathcal{H}_t | \mathbf{X}_{T-t}^{1:n})] \leq c_5 \frac{\bar{B}}{\sigma_{T-t}^2} \left(\frac{L^2 S \log(L S)}{n \log^{-1} n} \right)^{\frac{1}{2}},$$

where c_5 is an absolute constant. This completes the proof. $\square$

Theorem 4.3. Suppose Assumptions 1 and 2 hold. Let $t \in (0, T)$. Set the hypothesis class $\mathcal{H}_t$ as

$$\mathcal{H}_t := \left\{ h_t \in \text{conv}(N(L, S)) : \begin{array}{l} \sup_{\mathbf{x} \in \mathbb{R}^d} h_t(\mathbf{x}) \leq \bar{B}, \quad \inf_{\mathbf{x} \in \mathbb{R}^d} h_t(\mathbf{x}) \geq \underline{B}, \\ \max_{1 \leq k \leq d} \sup_{\mathbf{x} \in \mathbb{R}^d} |D_k h_t(\mathbf{x})| \leq 2\sigma_{T-t}^{-2} \bar{B}, \end{array} \right\}.$$

where $L = \mathcal{O}(\log n)$ and $S = \mathcal{O}(n^{\frac{d}{d+4}})$. Let $\hat{h}_t^\lambda$ be the gradient penalized empirical risk minimizer defined as (3.18), and let h_t^* be the Doob's h -function defined as (3.7). Then the following inequalities hold:

$$\mathbb{E} \left[\|\nabla \log \hat{h}_t^\lambda - \nabla \log h_t^*\|_{L^2(p_{T-t})}^2 \right] \leq C \sigma_{T-t}^{-8} n^{-\frac{1}{d+4}},$$

provided that the regularization parameter λ is set as $\lambda = \mathcal{O}(n^{-\frac{1}{d+4}})$, where C is a constant depending only on d , $\bar{B}$, and $\underline{B}$.

Proof of Theorem 4.3. Substituting Lemmas C.13, and C.14 into Lemma 4.2 yields

$$\begin{aligned} & \mathbb{E}[\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2] \\ & \lesssim \frac{\log^2 N}{\sigma_{T-t}^8 N^2} + \left(\frac{LS \log(S)}{n \log^{-1} n}\right)^{\frac{1}{2}} + \frac{\lambda}{\sigma_{T-t}^4} \left(\frac{L^2 S \log(LS)}{n \log^{-1} n}\right)^{\frac{1}{2}} + \frac{\lambda^2}{\sigma_{T-t}^8} \\ & \lesssim \frac{\log^2 N}{\sigma_{T-t}^8 N^2} + \left(\frac{N^d \log^2 N}{n \log^{-1} n}\right)^{\frac{1}{2}} + \frac{\lambda}{\sigma_{T-t}^4} \left(\frac{N^d \log^4 N}{n \log^{-1} n}\right)^{\frac{1}{2}} + \frac{\lambda^2}{\sigma_{T-t}^8}, \end{aligned}$$

where we omit the constants independent of t , N and n , and the second inequality is due to $L \leq C \log N$ and $S \leq N^d$. Similarly,

$$\begin{aligned} & \mathbb{E}[\|\nabla \hat{h}_t^\lambda - \nabla h_t^*\|_{L^2(p_{T-t})}^2] \\ & \lesssim \frac{1}{\lambda} \frac{\log^2 N}{\sigma_{T-t}^8 N^2} + \frac{1}{\lambda} \left(\frac{N^d \log^2 N}{n \log^{-1} n}\right)^{\frac{1}{2}} + \frac{1}{\sigma_{T-t}^4} \left(\frac{N^d \log^4 N}{n \log^{-1} n}\right)^{\frac{1}{2}} + \frac{\lambda}{\sigma_{T-t}^8}. \end{aligned}$$

By setting $N = n^{\frac{1}{d+4}}$ and $\lambda = n^{-\frac{1}{d+4}}$, we have

$$\begin{aligned} & \mathbb{E}[\|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2] \lesssim \frac{1}{\sigma_{T-t}^8} n^{-\frac{2}{d+4}} \log^2 n, \\ (C.31) \quad & \mathbb{E}[\|\nabla \hat{h}_t^\lambda - \nabla h_t^*\|_{L^2(p_{T-t})}^2] \lesssim \frac{1}{\sigma_{T-t}^8} n^{-\frac{1}{d+4}} \log^2 n. \end{aligned}$$

Consequently,

$$\begin{aligned} & \|\nabla \log \hat{h}_t^\lambda - \nabla \log h_t^*\|_{L^2(p_{T-t})}^2 \\ & = \left\| \frac{\nabla \hat{h}_t^\lambda}{\hat{h}_t^\lambda} - \frac{\nabla h_t^*}{h_t^*} + \frac{\nabla h_t^*}{\hat{h}_t^\lambda} - \frac{\nabla h_t^*}{h_t^*} \right\|_{L^2(p_{T-t})}^2 \\ & \leq 2 \left\| \frac{\nabla \hat{h}_t^\lambda}{\hat{h}_t^\lambda} - \frac{\nabla h_t^*}{\hat{h}_t^\lambda} \right\|_{L^2(p_{T-t})}^2 + 2 \left\| \frac{\nabla h_t^*}{\hat{h}_t^\lambda} - \frac{\nabla h_t^*}{h_t^*} \right\|_{L^2(p_{T-t})}^2 \\ & \leq \frac{2}{\bar{B}^2} \|\nabla \hat{h}_t^\lambda - \nabla h_t^*\|_{L^2(p_{T-t})}^2 + 2 \frac{\bar{B}^2}{\underline{B}^4} \|\hat{h}_t^\lambda - h_t^*\|_{L^2(p_{T-t})}^2 \\ & \lesssim \frac{1}{\sigma_{T-t}^8} n^{-\frac{1}{d+4}} \log^2 n, \end{aligned}$$

where the second inequality is owing to Lemma C.1, and the last inequality holds from (C.31). This completes the proof. $\square$

D Error Bounds of the Controllable Diffusion Models

D.1 Error decomposition of the controllable diffusion models.

Lemma 4.4. Suppose Assumptions 1, 2, and 3 hold. Let $\hat{q}_{T-T_0}$ be the marginal density of $\hat{\mathbf{Z}}_{T-T_0}^\leftarrow$ defined in (3.20). Then it follows that

$$\begin{aligned} \text{KL}(q_{T_0} \|\hat{q}_{T-T_0}) & \lesssim \underbrace{\frac{\bar{B}}{\underline{B}} \sum_{k=0}^{K-1} h \mathbb{E}^\mathbb{P} \left[\|\nabla \log \hat{h}_{kh}(\mathbf{X}_{kh}^\leftarrow) - \nabla \log h_{kh}^*(\mathbf{X}_{kh}^\leftarrow)\|_2^2 \right]}_{(I)} \\ & \quad + \underbrace{\frac{\bar{B}}{\underline{B}} T \varepsilon_{\text{base}}^2}_{(II)} + \underbrace{d \exp(-T)}_{(III)} + \underbrace{\frac{d^2 T^2}{\sigma_{T_0}^4 K}}_{(IV)}, \end{aligned}$$

where the notation $\lesssim$ hides absolute constants.

Proof of Lemma 4.4. According to Chen et al. (2023, Proposition C.3), we have

$$\begin{aligned}
 \text{KL}(q_{T_0} \parallel \hat{q}_{T-T_0}) &\lesssim \underbrace{\text{KL}(q_T \parallel \gamma_d)}_{\text{initial error}} + \underbrace{\sum_{k=0}^{K-1} h \mathbb{E}^{\mathbb{Q}} \left[\|\hat{\mathbf{s}}(kh, \mathbf{Z}_{kh}^{\leftarrow}) - \nabla \log p_{T-kh}(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2 \right]}_{\text{base score error}} \\
 &+ \underbrace{\sum_{k=0}^{K-1} h \mathbb{E}^{\mathbb{Q}} \left[\|\nabla \log \hat{h}_{kh}(\mathbf{Z}_{kh}^{\leftarrow}) - \nabla \log h_{kh}^*(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2 \right]}_{\text{Doob's guidance error}} \\
 &+ \underbrace{\sum_{k=0}^{K-1} \int_{kh}^{(k+1)h} \mathbb{E}^{\mathbb{Q}} \left[\|\nabla \log q_{T-t}(\mathbf{Z}_t^{\leftarrow}) - \nabla \log q_{T-kh}(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2 \right] dt}_{\text{discretization error}},
 \end{aligned}
 \tag{D.1}$$

where γ_d is the density of a standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.

Step 1. Initial error in (D.1). Using Chen et al. (2023, Lemma C.4), we have

$$\text{KL}(q_T \parallel \gamma_d) \lesssim d \exp(-T).
 \tag{D.2}$$

Step 2. Base score error and Doob's guidance error in (D.1). Under Assumption 2, it is apparent that $\mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow})] \geq \underline{B}$ and $\sup_{\mathbf{x} \in \mathbb{R}^d} h^*(t, \mathbf{x}) \leq \bar{B}$ for each $t \in (T_0, T)$. Hence, the density ratio is uniformly bounded

$$\sup_{\mathbf{x} \in \mathbb{R}^d} \frac{q_{T-t}(\mathbf{x})}{p_{T-t}(\mathbf{x})} = \frac{h^*(t, \mathbf{x})}{\mathbb{E}^{\mathbb{P}}[w(\mathbf{X}_T^{\leftarrow})]} \leq \frac{\bar{B}}{\underline{B}}.
 \tag{D.3}$$

For the base score error term in (D.1), it follows for each $0 \leq k \leq K-1$ that

$$\begin{aligned}
 &\mathbb{E}^{\mathbb{Q}} \left[\|\hat{\mathbf{s}}(kh, \mathbf{Z}_{kh}^{\leftarrow}) - \nabla \log p_{T-kh}(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2 \right] \\
 &= \int \|\hat{\mathbf{s}}(kh, \mathbf{z}) - \nabla \log p_{T-kh}(\mathbf{z})\|_2^2 q_{T-kh}(\mathbf{z}) d\mathbf{z} \\
 &= \int \|\hat{\mathbf{s}}(kh, \mathbf{x}) - \nabla \log p_{T-kh}(\mathbf{x})\|_2^2 \frac{q_{T-kh}(\mathbf{x})}{p_{T-kh}(\mathbf{x})} p_{T-kh}(\mathbf{x}) d\mathbf{x} \\
 &\leq \frac{\bar{B}}{\underline{B}} \int \|\hat{\mathbf{s}}(kh, \mathbf{x}) - \nabla \log p_{T-kh}(\mathbf{x})\|_2^2 p_{T-kh}(\mathbf{x}) d\mathbf{x} \\
 &= \frac{\bar{B}}{\underline{B}} \mathbb{E}^{\mathbb{P}} \left[\|\hat{\mathbf{s}}(kh, \mathbf{X}_{kh}^{\leftarrow}) - \nabla \log p_{T-kh}(\mathbf{X}_{kh}^{\leftarrow})\|_2^2 \right],
 \end{aligned}$$

where the inequality holds from (D.3) and Hölder's inequality. Consequently,

$$\begin{aligned}
 &\sum_{k=0}^{K-1} h \mathbb{E}^{\mathbb{Q}} \left[\|\hat{\mathbf{s}}(kh, \mathbf{Z}_{kh}^{\leftarrow}) - \nabla \log p_{T-kh}(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2 \right] \\
 &\leq \frac{\bar{B}}{\underline{B}} \sum_{k=0}^{K-1} h \mathbb{E}^{\mathbb{P}} \left[\|\hat{\mathbf{s}}(kh, \mathbf{X}_{kh}^{\leftarrow}) - \nabla \log p_{T-kh}(\mathbf{X}_{kh}^{\leftarrow})\|_2^2 \right] \leq T \frac{\bar{B}}{\underline{B}} \varepsilon_{\text{base}}^2,
 \end{aligned}
 \tag{D.4}$$

where the last inequality is owing to Assumption 3. By a similar argument, we have

$$\begin{aligned}
 & \sum_{k=0}^{K-1} h \mathbb{E}^{\mathbb{Q}} \left[\|\nabla \log \hat{h}_{kh}(\mathbf{Z}_{kh}^{\leftarrow}) - \nabla \log h_{kh}^*(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2 \right] \\
 (D.5) \quad & \leq \frac{\bar{B}}{\underline{B}} \sum_{k=0}^{K-1} h \mathbb{E}^{\mathbb{P}} \left[\|\nabla \log \hat{h}_{kh}(\mathbf{X}_{kh}^{\leftarrow}) - \nabla \log h_{kh}^*(\mathbf{X}_{kh}^{\leftarrow})\|_2^2 \right].
 \end{aligned}$$

Step 3. Discretization error in (D.1). According to Chen et al. (2023, Lemma D.1), we have

$$(D.6) \quad \mathbb{E}^{\mathbb{Q}} [\|\nabla \log q_{T-t}(\mathbf{Z}_t^{\leftarrow}) - \nabla \log q_{T-kh}(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2] \lesssim \frac{dG_k T}{K},$$

for any $t \in (kh, (k+1)h)$, provided that $\nabla \log q_{T-t}$ is G -Lipschitz for any $t \in (kh, (k+1)h)$. Then it remains to estimate the Lipschitz constant G_k . Using Lemmas C.2 and C.3 yields

$$\begin{aligned}
 \nabla^2 \log q_{T-t}(\mathbf{z}) &= -\frac{1}{\sigma_{T-t}^2} \mathbf{I}_d + \frac{\mu_{T-t}}{\sigma_{T-t}^2} \nabla \mathbb{E}[\mathbf{Z}_0 | \mathbf{Z}_t = \mathbf{z}] \\
 &= -\frac{1}{\sigma_{T-t}^2} \mathbf{I}_d + \frac{\mu_{T-t}^2}{\sigma_{T-t}^4} \text{Cov}(\mathbf{Z}_0 | \mathbf{Z}_t = \mathbf{z}).
 \end{aligned}$$

As a consequence, for each $0 \leq k \leq K-1$,

$$\begin{aligned}
 G_k &\leq \sup_{t \in (T_0, T)} \sup_{\mathbf{z} \in \mathbb{R}^d} \|\nabla^2 \log q_{T-t}(\mathbf{z})\|_{\text{op}} \\
 &\leq \sup_{t \in (T_0, T)} \frac{1}{\sigma_{T-t}^2} + \frac{\mu_{T-t}^2}{\sigma_{T-t}^4} \|\text{Cov}(\mathbf{Z}_0 | \mathbf{Z}_t = \mathbf{z})\|_{\text{op}} \\
 (D.7) \quad &\lesssim \frac{d}{\sigma_{T_0}^4},
 \end{aligned}$$

where the first inequality holds from the triangular inequality, and the second inequality is due to the boundedness of $\mathbf{Z}_0$ under Assumptions 1 and 2. Combining (D.6) and (D.7) implies

$$(D.8) \quad \sum_{k=0}^{K-1} \int_{kh}^{(k+1)h} \mathbb{E}^{\mathbb{Q}} [\|\nabla \log q_{T-t}(\mathbf{Z}_t^{\leftarrow}) - \nabla \log q_{T-kh}(\mathbf{Z}_{kh}^{\leftarrow})\|_2^2] dt \lesssim \frac{d^2 T^2}{\sigma_{T_0}^4 K}.$$

Step 4. Conclusions. Substituting (D.2), (D.4), (D.5), (D.8) into (D.1) completes the proof. $\square$

Corollary D.1. Suppose Assumptions 1, 2, and 3 hold. Let $\delta \in (0, 1)$. Set the hypothesis classes $\{\mathcal{H}_{T-kh}\}_{k=0}^{K-1}$ as (4.3) with the same depth L and number of non-zero parameters S . Let $\hat{q}_{T-T_0}$ be the marginal density of $\hat{\mathbf{Z}}_{T-T_0}^{\leftarrow}$ defined in (3.20). Then it follows that

$$\|\hat{q}_{T-T_0} - \hat{q}_{T-T_0}\|_{\text{TV}}^2 \leq \frac{C\delta^2}{\sigma_{T_0}^8},$$

where C is a constant depending only on d , $\bar{B}$, and $\underline{B}$, and

$$\begin{aligned}
 K &= \Theta\left(\frac{\sigma_{T_0}^4}{\delta^2} \log^2\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \quad T = \Theta\left(\log\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \quad h = \frac{T}{K} = \Theta\left(\frac{\delta^2}{\sigma_{T_0}^4} \log^{-1}\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \\
 \varepsilon_{\text{base}}^2 &= \Theta\left(\frac{\delta^2}{\sigma_{T_0}^8} \log^{-1}\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \quad n = \Theta\left(\frac{1}{\delta^{2(d+4)}} \log^{d+4}\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right),
 \end{aligned}$$

Proof of Corollary D.1. Combining Theorem 4.3 and Lemma 4.4 yields

$$\text{KL}(q_{T_0} \|\hat{q}_{T-T_0}) \lesssim C' \frac{\bar{B}T}{\underline{B}\sigma_{T_0}^8} n^{-\frac{1}{d+4}} + \frac{\bar{B}}{\underline{B}} T \varepsilon_{\text{base}}^2 + d \exp(-T) + \frac{d^2 T^2}{\sigma_{T_0}^4 K},$$

where C' is a constant depending only on d , $\bar{B}$, and $\underline{B}$. By setting

$$\begin{aligned} K &= \Theta\left(\frac{\sigma_{T_0}^4}{\delta^2} \log^2\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \quad T = \Theta\left(\log\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \\ \varepsilon_{\text{base}}^2 &= \Theta\left(\frac{\delta^2}{\sigma_{T_0}^8} \log^{-1}\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \quad n = \Theta\left(\frac{1}{\delta^{2(d+4)}} \log^{d+4}\left(\frac{\sigma_{T_0}^8}{\delta^2}\right)\right), \end{aligned}$$

we find

$$\text{KL}(q_{T_0} \|\hat{q}_{T-T_0}) \leq \frac{C\delta^2}{\sigma_{T_0}^8}.$$

Finally, using Pinker's inequality completes the proof. $\square$

D.2 Convergence rate of the controllable diffusion models.

Theorem 4.5. Suppose Assumptions 1, 2, and 3 hold. Let $\varepsilon \in (0, 1)$. Set the hypothesis classes $\{\mathcal{H}_{T-kh}\}_{k=0}^{K=1}$ as (4.3) with the same depth L and number of non-zero parameters S as Theorem 4.3. Let $\hat{q}_{T-T_0}$ be the marginal density of $\hat{\mathbf{Z}}_{T-T_0}^{\leftarrow}$ defined in (3.20), and let $(\mathcal{M} \circ \mathcal{T}_R)_{\#} \hat{q}_{T-T_0}$ defined as (3.21). Then it follows that

$$\mathbb{E}\left[\mathcal{W}_2^2(q_0, (\mathcal{M} \circ \mathcal{T}_R)_{\#} \hat{q}_{T-T_0})\right] \leq C\left\{\sigma_{T_0}^2 + \varepsilon + \frac{\varepsilon^3}{\sigma_{T_0}^4} \log\left(\frac{1}{\varepsilon}\right)\right\},$$

provided that the truncation radius R , the step size h , the number of steps K , the error of base score $\varepsilon_{\text{base}}$, and the number of samples n for Doob's matching are set, respectively, as

$$\begin{aligned} R &= \Theta\left(\log^{\frac{1}{2}}\left(\frac{1}{\varepsilon}\right)\right), \quad h = \Theta\left(\varepsilon^4 \log^{-1}\left(\frac{1}{\varepsilon}\right)\right), \quad K = \Theta\left(\frac{1}{\varepsilon^4} \log^2\left(\frac{1}{\varepsilon}\right)\right), \\ \varepsilon_{\text{base}}^2 &= \Theta\left(\varepsilon^2 \log^{-1}\left(\frac{1}{\varepsilon}\right)\right), \quad n = \Theta\left(\frac{1}{\varepsilon^{6(d+4)}} \log^{d+4}\left(\frac{1}{\varepsilon}\right)\right), \end{aligned}$$

Further, letting $T_0 = \Theta(\varepsilon)$, it holds that

$$\mathbb{E}\left[\mathcal{W}_2^2(q_0, (\mathcal{M} \circ \mathcal{T}_R)_{\#} \hat{q}_{T-T_0})\right] \leq C\varepsilon \log\left(\frac{1}{\varepsilon}\right).$$

Here C is a constant depending only on d , $\bar{B}$, and $\underline{B}$.

Proof of Theorem 4.5. According to the triangular inequality, we have

$$\begin{aligned} \mathcal{W}_2^2(q_0, (\mathcal{M} \circ \mathcal{T}_R)_{\#} \hat{q}_{T-T_0}) &= \underbrace{3\mathcal{W}_2^2(q_0, \mathcal{M}_{\#} q_{T_0})}_{(i)} + \underbrace{3\mathcal{W}_2^2(\mathcal{M}_{\#} q_{T_0}, (\mathcal{M} \circ \mathcal{T}_R)_{\#} q_{T_0})}_{(ii)} \\ (D.9) \quad &+ \underbrace{3\mathcal{W}_2^2((\mathcal{M} \circ \mathcal{T}_R)_{\#} q_{T_0}, (\mathcal{M} \circ \mathcal{T}_R)_{\#} \hat{q}_{T-T_0})}_{(iii)}. \end{aligned}$$

Here the term (i) represents the early-stopping error, the term (ii) represents the truncation error, while the term (iii) represents the error of controllable diffusion models (3.20). In the rest of the proof, we bound these three errors, respectively.

Step 1. Bound the term (i) in (D.9). To estimate the 2-Wasserstein distance between the target distribution q_0 and the scaled early-stopping distribution $\mathcal{M}_{\sharp} q_{T_0}$, we begin by producing a coupling of them. Let $\mathbf{Z}_0 \sim q_0$, and let $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ be independent of $\mathbf{Z}_0$. Define $\tilde{\mathbf{Z}}_{T_0} := \mathbf{Z}_0 + \sigma_{T_0} \mu_{T_0}^{-1} \varepsilon$. It is apparent that $\tilde{\mathbf{Z}}_{T_0} \sim \mathcal{M}_{\sharp} q_{T_0}$. Then

$$(D.10) \quad \mathcal{W}_2^2(q_0, \mathcal{M}_{\sharp} q_{T_0}) \leq \mathbb{E}[\|\mathbf{Z}_0 - \tilde{\mathbf{Z}}_{T_0}\|_2^2] = \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} \mathbb{E}[\|\varepsilon\|_2^2] = \frac{d\sigma_{T_0}^2}{\mu_{T_0}^2}.$$

Step 2. Bound the term (ii) in (D.9). Let $\mathbf{Z}_{T_0} \sim q_{T_0}$. According to the definition of the truncation operator $\mathcal{T}_R$, the joint law of $(\mu_{T_0}^{-1} \mathbf{Z}_{T_0}, \mu_{T_0}^{-1} \mathbf{Z}_{T_0} \mathbb{1}_{B(\mathbf{0}, R)}(\mathbf{Z}_{T_0}))$ is a coupling of $\mathcal{M}_{\sharp} q_{T_0}$ and $(\mathcal{M} \circ \mathcal{T}_R)_{\sharp} q_{T_0}$. Therefore,

$$\begin{aligned} \mathcal{W}_2^2(\mathcal{M}_{\sharp} q_{T_0}, (\mathcal{M} \circ \mathcal{T}_R)_{\sharp} q_{T_0}) &\leq \mathbb{E}[\|\mu_{T_0}^{-1} \mathbf{Z}_{T_0} - \mu_{T_0}^{-1} \mathbf{Z}_{T_0} \mathbb{1}_{B(\mathbf{0}, R)}(\mathbf{Z}_{T_0})\|_2^2] \\ &= \frac{1}{\mu_{T_0}^2} \int \|\mathbf{z} - \mathbf{z} \mathbb{1}_{B(\mathbf{0}, R)}(\mathbf{z})\|_2^2 q_{T_0}(\mathbf{z}) d\mathbf{z} \\ &= \frac{1}{\mu_{T_0}^2} \int \|\mathbf{z}\|_2^2 \mathbb{1}_{\mathbb{R}^d - B(\mathbf{0}, R)}(\mathbf{z}) q_{T_0}(\mathbf{z}) d\mathbf{z} \\ &\leq \frac{1}{\mu_{T_0}^2} \mathbb{E}^{\frac{1}{2}}[\|\mathbf{Z}_{T_0}\|_2^4] \Pr^{\frac{1}{2}}\{\|\mathbf{Z}_{T_0}\|_2 > R\} \\ &\lesssim \frac{1}{\mu_{T_0}^2} d 2^{d+1} \exp\left(-\frac{R^2}{4d\mu_{T_0}^2 + 8\sigma_{T_0}^2}\right), \end{aligned}$$

where the second ineq holds from Cauchy-Schwarz inequality, and the last inequality is due to Lemma E.3 and Corollary E.2. By setting $R^2 = (4d\mu_{T_0}^2 + 8\sigma_{T_0}^2) \log(\varepsilon^{-1})$, we have

$$(D.11) \quad \mathcal{W}_2^2(\mathcal{M}_{\sharp} q_{T_0}, (\mathcal{M} \circ \mathcal{T}_R)_{\sharp} q_{T_0}) \lesssim \frac{d 2^d}{\mu_{T_0}^2} \varepsilon.$$

Step 3. Bound the term (iii) in (D.9). Let $\mathbf{Z}_{T_0}^R \sim (\mathcal{T}_R)_{\sharp} q_{T_0}$ and $\hat{\mathbf{Z}}_{T_0}^R \sim (\mathcal{T}_R)_{\sharp} \hat{q}_{T-T_0}$ be optimal coupled. This means

$$(D.12) \quad \mathcal{W}_2^2((\mathcal{T}_R)_{\sharp} q_{T_0}, (\mathcal{T}_R)_{\sharp} \hat{q}_{T-T_0}) = \mathbb{E}[\|\mathbf{Z}_{T_0}^R - \hat{\mathbf{Z}}_{T_0}^R\|_2^2].$$

On the other hand, $\mu_{T_0}^{-1} \mathbf{Z}_{T_0}^R \sim (\mathcal{M} \circ \mathcal{T}_R)_{\sharp} q_{T_0}$ and $\mu_{T_0}^{-1} \hat{\mathbf{Z}}_{T_0}^R \sim (\mathcal{M} \circ \mathcal{T}_R)_{\sharp} \hat{q}_{T-T_0}$. Hence,

$$(D.13) \quad \begin{aligned} &\mathcal{W}_2^2((\mathcal{M} \circ \mathcal{T}_R)_{\sharp} q_{T_0}, (\mathcal{M} \circ \mathcal{T}_R)_{\sharp} \hat{q}_{T-T_0}) \\ &\leq \mathbb{E}[\|\mu_{T_0}^{-1} \mathbf{Z}_{T_0}^R - \mu_{T_0}^{-1} \hat{\mathbf{Z}}_{T_0}^R\|_2^2] = \frac{1}{\mu_{T_0}^2} \mathcal{W}_2^2((\mathcal{T}_R)_{\sharp} q_{T_0}, (\mathcal{T}_R)_{\sharp} \hat{q}_{T-T_0}), \end{aligned}$$

where the equality holds from (D.12). Then using Villani (2009, Theorem 6.15) and the data processing inequality, we have

$$(D.14) \quad \begin{aligned} \mathcal{W}_2^2((\mathcal{T}_R)_{\sharp} q_{T_0}, (\mathcal{T}_R)_{\sharp} \hat{q}_{T-T_0}) &= 2R^2 \|(\mathcal{T}_R)_{\sharp} q_{T_0} - (\mathcal{T}_R)_{\sharp} \hat{q}_{T-T_0}\|_{\text{TV}} \\ &\leq 2R^2 \|q_{T_0} - \hat{q}_{T-T_0}\|_{\text{TV}}. \end{aligned}$$

Combining (D.13) and (D.14) yields

$$\begin{aligned}
 \mathcal{W}_2^2((\mathcal{M} \circ \mathcal{T}_R)_\# q_{T_0}, (\mathcal{M} \circ \mathcal{T}_R)_\# \hat{q}_{T-T_0}) &\leq \frac{2R^2}{\mu_{T_0}^2} \|q_{T_0} - \hat{q}_{T-T_0}\|_{\text{TV}} \\
 &\leq \frac{2R^2}{\mu_{T_0}^2} \frac{C' \varepsilon^3}{\sigma_{T_0}^4} \\
 &\leq \frac{2(4d\mu_{T_0}^2 + 8\sigma_{T_0}^2) \log(\varepsilon^{-1})}{\mu_{T_0}^2} \frac{C' \varepsilon^3}{\sigma_{T_0}^4},
 \end{aligned}
 \tag{D.15}$$

where C is a constant depending only on d , $\bar{B}$, and $\underline{B}$, and the second inequality holds from Corollary D.1 with $\delta = \varepsilon^3$.

Step 4. Conclusion. Substituting (D.10), (D.11), and (D.15) into (D.9) yields

$$\begin{aligned}
 \mathcal{W}_2^2(q_0, (\mathcal{M} \circ \mathcal{T}_R)_\# \hat{q}_{T-T_0}) &\lesssim \frac{d\sigma_{T_0}^2}{\mu_{T_0}^2} + \frac{d2^d}{\mu_{T_0}^2} \varepsilon + \frac{2(4d\mu_{T_0}^2 + 8\sigma_{T_0}^2) \log(\varepsilon^{-1})}{\mu_{T_0}^2} \frac{C' \varepsilon^3}{\sigma_{T_0}^4} \\
 &\leq C \left\{ \sigma_{T_0}^2 + \varepsilon + \frac{\varepsilon^3}{\sigma_{T_0}^4} \log\left(\frac{1}{\varepsilon}\right) \right\},
 \end{aligned}$$

where C is a constant depending only on d , $\bar{B}$, and $\underline{B}$. Letting $\sigma_{T_0}^2 = \Theta(\varepsilon)$, i.e., $T_0 = \Theta(\varepsilon)$, completes the proof. $\square$

E Auxiliary Lemmas

Lemma E.1. *Suppose Assumption 1 holds. Let $\mathbf{X}_t \sim p_t$. Then for each $\xi > 0$,*

$$\Pr \{ \|\mathbf{X}_t\|_2 \geq \xi \} \leq 2^{d+1} \exp \left(- \frac{\xi^2}{4d\mu_t^2 + 8\sigma_t^2} \right).$$

Proof of Lemma E.1. According to Assumption 1, we have

$$\mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0\|_2^2}{2d\mu_t^2} \right) \right] \leq 2.
 \tag{E.1}$$

Let $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_d)$. Then it follows that

$$\begin{aligned}
 \mathbb{E} \left[\exp \left(\frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{4\sigma_t^2} \right) \right] &= \mathbb{E} \left[\exp \left(\frac{\|\boldsymbol{\varepsilon}\|_2^2}{4} \right) \right] \\
 &= (2\pi)^{-\frac{d}{2}} \int \exp \left(\frac{\|\boldsymbol{\varepsilon}\|_2^2}{4} \right) \exp \left(- \frac{\|\boldsymbol{\varepsilon}\|_2^2}{2} \right) d\boldsymbol{\varepsilon} \\
 &= (2\pi)^{-\frac{d}{2}} \int \exp \left(- \frac{\|\boldsymbol{\varepsilon}\|_2^2}{4} \right) d\boldsymbol{\varepsilon} \leq 2^d.
 \end{aligned}
 \tag{E.2}$$

Notice that $\mathbf{X}_t \stackrel{d}{=} \mu_t \mathbf{X}_0 + \sigma_t \boldsymbol{\varepsilon}$, where $\mathbf{X}_0 \sim p_0$ and $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_d)$ are independent. Therefore,

$$\begin{aligned}
 \mathbb{E} \left[\exp \left(\frac{\|\mathbf{X}_t\|_2^2}{4d\mu_t^2 + 8\sigma_t^2} \right) \right] &= \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0 + \sigma_t \boldsymbol{\varepsilon}\|_2^2}{4d\mu_t^2 + 8\sigma_t^2} \right) \right] \\
 &\leq \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0\|_2^2}{2d\mu_t^2 + 4\sigma_t^2} + \frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{2d\mu_t^2 + 4\sigma_t^2} \right) \right] \\
 &\leq \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0\|_2^2}{2d\mu_t^2 + 4\sigma_t^2} \right) \right] \mathbb{E} \left[\exp \left(\frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{2d\mu_t^2 + 4\sigma_t^2} \right) \right] \\
 &\leq \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0\|_2^2}{2d\mu_t^2} \right) \right] \mathbb{E} \left[\exp \left(\frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{4\sigma_t^2} \right) \right] \leq 2^{d+1},
 \end{aligned}
 \tag{E.3}$$

where the first inequality follows from Cauchy-Schwarz inequality, the second inequality holds from the independence of $\mathbf{X}_0$ and $\boldsymbol{\varepsilon}$, and the last inequality is due to (E.1) and (E.2). Then we aim to bound the tail probability. For each $\xi > 0$, we have

$$\begin{aligned} \Pr \{ \|\mathbf{X}_t\|_2 \geq \xi \} &= \Pr \left\{ \frac{\|\mathbf{X}_t\|_2^2}{4d\mu_t^2 + 8\sigma_t^2} \geq \frac{\xi^2}{4d\mu_t^2 + 8\sigma_t^2} \right\} \\ &= \Pr \left\{ \exp \left(\frac{\|\mathbf{X}_t\|_2^2}{4d\mu_t^2 + 8\sigma_t^2} \right) \geq \exp \left(\frac{\xi^2}{4d\mu_t^2 + 8\sigma_t^2} \right) \right\} \\ &\leq \exp \left(- \frac{\xi^2}{4d\mu_t^2 + 8\sigma_t^2} \right) \mathbb{E} \left[\exp \left(\frac{\|\mathbf{X}_t\|_2^2}{4d\mu_t^2 + 8\sigma_t^2} \right) \right] \\ &\leq 2^{d+1} \exp \left(- \frac{\xi^2}{4d\mu_t^2 + 8\sigma_t^2} \right), \end{aligned}$$

where the first inequality invokes Markov's inequality, and the last inequality is due to (E.3). This completes the proof. $\square$

Corollary E.2. *Suppose Assumptions 1 and 2 hold. Let $\mathbf{Z}_t \sim q_t$. Then for each $\xi > 0$,*

$$\Pr \{ \|\mathbf{Z}_t\|_2 \geq \xi \} \leq 2^{d+1} \exp \left(- \frac{\xi^2}{4d\mu_t^2 + 8\sigma_t^2} \right).$$

Proof of Corollary E.2. Under Assumptions 1 and 2, $\text{supp}(q_0) = \text{supp}(p_0)$. The same argument as Lemma E.1 completes the proof. $\square$

Lemma E.3. *Suppose Assumptions 1 and 2 hold. Let $\mathbf{Z}_t \sim q_t$. Then for each $\xi > 0$,*

$$\mathbb{E} [\|\mathbf{Z}_t\|_2^4] \lesssim d^2.$$

Proof of Lemma E.3. Let $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_d)$. It is straightforward that

$$\mathbb{E} [\|\boldsymbol{\varepsilon}\|_2^4] = 4\Gamma\left(\frac{d+4}{2}\right)\Gamma\left(\frac{d}{2}\right) \leq (d+4)^2.$$

Since $\mathbf{Z}_t \stackrel{d}{=} \mu_t \mathbf{Z}_0 + \sigma_t \boldsymbol{\varepsilon}$ with $\mathbf{Z}_0 \sim q_0$ independent of $\boldsymbol{\varepsilon}$, it follows from the triangular inequality that

$$\begin{aligned} \mathbb{E} [\|\mathbf{Z}_t\|_2^4] &\leq 8\mu_t^4 \mathbb{E} [\|\mathbf{Z}_0\|_2^4] + 8\sigma_t^4 \mathbb{E} [\|\boldsymbol{\varepsilon}\|_2^4] \\ &\leq 8(d^2 + (d+4)^2), \end{aligned}$$

where we used the fact that $\mu_t, \sigma_t \leq 1$, and $\text{supp}(q_0) = \text{supp}(p_0)$ under Assumptions 1 and 2. This completes the proof. $\square$